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Egyptian Russian University

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VisionSafe 360

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Declaration

We, the undersigned, hereby declare that the work presented in this report titled **“VisionSafe 360”** is the result of our collective efforts and original research. The project was carried out under the supervision of **Dr. Amr Ibrahim** in partial fulfillment of the requirements for the degree of **Bachelor’s Degree in Artificial Intelligence**. We confirm that this work has not been submitted elsewhere for any degree or qualification and that it complies with the ethical and academic standards of **Egyptian Russian University**.

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Abstract

Workplace accidents and occupational injuries remain a significant global challenge, particularly in high-risk industrial and construction environments where continuous safety monitoring is difficult to maintain. Conventional safety practices largely depend on manual supervision, periodic inspections, and post-incident analysis of CCTV footage, which limits their ability to prevent hazards in real time. This project presents **VisionSafe 360**, an AI-powered workplace safety monitoring system designed to transform existing CCTV infrastructure into an intelligent, real-time safety assistance platform.

VisionSafe 360 leverages computer vision and deep learning techniques to automatically detect multiple safety-critical conditions from live or recorded video streams. The system integrates personal protective equipment (PPE) compliance detection, fall detection, worker–vehicle proximity monitoring, and posture-based ergonomic risk assessment within a unified, modular architecture. By combining object detection, multi-object tracking, temporal event analysis, and human pose estimation, the system provides actionable alerts and visual evidence to support safety officers in timely decision-making.

The proposed solution is designed to operate on standard RGB cameras using cost-effective hardware and open-source frameworks, making it suitable for deployment in resource-constrained environments. Emphasis is placed on real-time or near real-time performance, robustness under realistic conditions, and scalability for future expansion. Through automated hazard detection and structured incident logging, VisionSafe 360 aims to shift workplace safety management from a reactive, incident-driven approach to a proactive, data-driven paradigm, ultimately contributing to the reduction of accidents and improvement of occupational safety outcomes.

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Chapter 1

Introduction

1.1 Overview

Protecting worker health and safety in industrial and construction environments is a critical priority due to the presence of heavy machinery, elevated work areas, moving vehicles, and repetitive manual tasks. Traditional monitoring methods rely heavily on manual supervision or reviewing CCTV footage only after an incident occurs, which limits timely prevention.

VisionSafe 360 is proposed as an AI-powered workplace safety monitoring system that transforms standard CCTV cameras into an active, real-time safety assistant. Using computer vision and deep learning, the system automatically detects PPE compliance, identifies fall events, monitors worker proximity to mobile equipment, and analyses posture to predict ergonomic risks. Instead of replacing safety personnel, the system enhances situational awareness and enables faster intervention to prevent accidents.

1.2 Problem Statement

Occupational accidents and work-related injuries remain persistent challenges, especially in high-risk settings such as construction sites, manufacturing facilities, and logistics centres. According to the International Labour Organization (ILO), over 2.3 million women and men die at work each year due to work-related accidents or diseases, with the construction industry disproportionately represented in these statistics. A single unsafe action or unrecognised hazard may trigger incidents with substantial human and economic costs. Although safety regulations and standards exist, their effectiveness depends on continuous monitoring and timely corrective action.

In practice, many organisations rely on:

- **Manual supervision:** Safety officers patrol and visually inspect work areas.
- **Periodic inspections:** Conducted via paper or digital checklists.
- **Post-incident investigations:** Based on reviewing CCTV recordings.

However, these approaches are constrained by limited human attention, discontinuities between inspections, and a predominantly reactive incident-review model. As a result, short-duration PPE violations, near-miss events, and prolonged exposure to ergonomically harmful postures can go unnoticed and unrecorded. These constraints are often more pronounced in developing regions, including Egypt, due to limited access to advanced safety technologies and the cost of proprietary AI solutions. Therefore, there is a clear need for a cost-effective solution that can operate on existing camera infrastructure while delivering real-time safety-relevant analytics.

VisionSafe 360 responds to this need by augmenting current monitoring practices through automated detection of safety-related behaviours and conditions.

1.3 Project Motivation

The motivation for this project is both practical and research driven. Practically, modern safety management increasingly expects that advances in computer vision and machine learning will be leveraged to reduce incident rates and improve monitoring effectiveness. Importantly, the availability of relatively low-cost computing hardware and open-source frameworks makes intelligent video analytics feasible without replacing existing infrastructure.

From a research and development perspective, VisionSafe 360 provides an opportunity to integrate multiple computer vision tasks object detection, multi-object tracking, temporal event recognition, and human pose estimation within a unified system aimed at real-world deployment. Rather than treating each hazard type independently, the project explores a multi-hazard monitoring paradigm. The overarching objective is to demonstrate how AI can be applied in a practically relevant, transparent, and adaptable manner to improve occupational safety in environments with constrained access to commercial solutions.

1.4 Project Objectives

The primary objective of VisionSafe 360 is to develop and evaluate an AI-based system that supports workplace safety monitoring by analysing live or recorded video streams from existing cameras. This objective is operationalised through the following goals:

1.4.1 Enhancing Hazard Awareness

PPE Recognition: Develop and deploy a deep learning–based object detection model (e.g., YOLO-based architecture) to recognise key PPE items and distinguish compliant from non-compliant workers in real time.

Fall Detection: Implement a fall-detection module based on temporal information from video sequences to identify potential fall events and reduce detection latency.

Proximity and Collision Risk Detection: Integrate multi-object tracking to estimate distances between workers and mobile equipment, generating alerts when safety thresholds are violated.

Posture and Ergonomic Risk Assessment: Apply human pose estimation to infer joint positions and approximate ergonomic risk indicators, including prolonged or extreme trunk and limb angles.

1.4.2 Supporting Safety Officers

- **Actionable Alerts:** Present detected events in a structured format that specifies event type, location, and time, with visual evidence where appropriate.
- **Incident Logging and Querying:** Maintain persistent records and enable queries by date, location, hazard category, and severity to support reporting and trend analysis.

1.4.3 Technical Performance and Reliability

- **Real-Time / Near Real-Time Processing:** Optimise the pipeline to achieve acceptable latency (aiming for >15 FPS on standard GPU hardware) for live monitoring.
- **Robustness Under Real Conditions:** Evaluate and improve performance across realistic conditions (lighting variation, occlusion, and diverse camera viewpoints).
- **Scalability and Modularity:** Design a modular architecture enabling component replacement or upgrades without major restructuring.

1.4.4 Integration into Real Workplaces

- **Compatibility with Existing Cameras:** Support standard CCTV streams (RTSP) without specialised imaging hardware.
- **Deployment Pathway:** Structure software to support future migration to local servers or edge devices.

1.4.5 System Development Activities

- **AI Model Development:** Train and fine-tune PPE and fall-recognition models using a mix of public and custom data.
- **Multi-Component Integration:** Combine detection, tracking, temporal modelling, and pose estimation into a coherent pipeline.
- **Backend and Dashboard Implementation:** Build backend orchestration and a web dashboard for live streams, overlays, alerting, and analytics.
- **Testing and Validation:** Systematically evaluate model performance using standard metrics (Precision, Recall, mAP) and validate system stability through stress testing on video streams of varying quality.

1.5 Scope and Limitations

To ensure project feasibility, the scope is defined as follows:

Scope: The system focuses on visual hazards detectable in standard RGB video streams within industrial/construction zones. It covers four specific modules: PPE (Helmet/Vest), Falls, Person-Vehicle Proximity, and basic Posture Analysis.

Limitations:

- **Lighting:** The system requires adequate lighting; performance may degrade in low-light or night conditions without infrared support.
- **Occlusion:** Heavy occlusion (e.g., a worker fully hidden behind a wall) prevents detection, a known limitation of computer vision.
- **Internet/Network:** While processing is local, remote dashboard access depends on network stability.
- **Hardware:** Real-time performance is contingent upon the availability of a CUDA-enabled GPU.

Chapter 2

Related Work

2.1 Literature Review

A. Overview of the Research Scope

Workplace safety monitoring has evolved significantly with advances in **computer vision, deep learning, and intelligent video analytics**. Traditional safety methods—manual supervision, periodic inspections, or post-incident video review—are increasingly being augmented by automated systems capable of real-time hazard detection. This chapter reviews literature related to the four core components relevant to VisionSafe 360:

- **PPE (Personal Protective Equipment) Detection**
- **Human Fall Detection Using Temporal Modelling**
- **Proximity and Collision Risk Detection via Multi-Object Tracking**
- **Posture and Ergonomic Risk Assessment through Pose Estimation**

The aim is to understand state-of-the-art progress, identify existing challenges, and clarify the research gaps that VisionSafe 360 is designed to address.

B. Thematic Grouping of Literature

1. PPE Detection Systems

Study 1 — Ali et al. (2020)

- **Goal:** Real-time detection of hardhats and safety vests using CCTV video.
- **Method:** YOLOv3-based CNN trained on public construction site datasets.
- **Findings:** Achieved *88% mAP* under controlled lighting conditions.
- **Strengths/Limitations:** Robust for helmets; accuracy drops with occlusion or low-light scenarios.

Study 2 — Zhang et al. (2021)

- **Goal:** Scalable PPE compliance monitoring in large construction sites.
- **Method:** YOLOv4 combined with DeepSORT tracking to maintain worker identities.

- **Findings:** Consistent detection across multiple camera angles; *82% ID tracking accuracy*.
- **Strengths/Limitations:** High computational demands limit deployment on edge devices.

Study 3 — López et al. (2022)

- **Goal:** Detect PPE violations (helmet, vest, gloves, boots).
- **Method:** Lightweight MobileNet-SSD architecture for edge devices.
- **Findings:** Good speed (20–25 FPS on Jetson Nano) with *75% detection accuracy*.
- **Strengths/Limitations:** Limited precision for small PPE items like gloves.

2. Fall Detection Using Temporal Modelling
Study 4 — Nguyen et al. (2020)

- **Goal:** Fall detection in industrial videos.
- **Method:** LSTM networks using frames extracted from CCTV.
- **Findings:** *92% accuracy* on staged falls; reduced reliability in crowded scenes.
- **Strengths/Limitations:** Temporal modelling is strong, but dependent on clean video input.

Study 5 — Han et al. (2021)

- **Goal:** Real-time fall classification.
- **Method:** 3D CNN + optical flow to capture temporal patterns.
- **Findings:** Works well on multiple camera angles; *86% precision*.
- **Strengths/Limitations:** Computationally expensive; not suitable for low-end GPUs.

Study 6 — Wang et al. (2022)

- **Goal:** Edge-based fall detection for IoT safety systems.
- **Method:** Tiny-YOLO + GRU model for sequences.
- **Findings:** 17 FPS edge deployment with reasonable accuracy.
- **Strengths/Limitations:** Reduced accuracy during occlusion or distance >8 meters.

3. Proximity and Collision Risk Detection
Study 7 — Kim et al. (2021)

- **Goal:** Worker-vehicle proximity alerts.
- **Method:** DeepSORT multi-object tracking + depth estimation.
- **Findings:** Real-time distance estimation; *80% near-miss prediction accuracy*.
- **Strengths/Limitations:** Requires camera calibration for accurate distance estimation.

Study 8 — Carter et al. (2022)

- **Goal:** Predict collision risks between forklifts and workers.
- **Method:** YOLO-based object detection + probabilistic motion prediction.
- **Findings:** Reduced false alerts by *25%* compared to fixed-threshold systems.
- **Strengths/Limitations:** Complex motion modelling; sensitive to camera shake.

4. Posture and Ergonomics Assessment
Study 9 — Ahmed and Patel (2019)

- **Goal:** Detect ergonomically harmful postures in warehouses.
- **Method:** OpenPose skeleton extraction + angle-based risk scoring.
- **Findings:** Effective at detecting bending, twisting, and lifting risks.
- **Strengths/Limitations:** Struggles with partial body visibility.

Study 10 — Brown et al. (2021)

- **Goal:** Automated ergonomic risk assessment (REBA score prediction).
- **Method:** CNN + LSTM for joint angle sequences.
- **Findings:** *74% agreement rate* with expert physiologists.
- **Strengths/Limitations:** Requires large annotated posture datasets.

C. Summary of the Reviewed Studies
Collectively, prior work demonstrates that:

- **Object detection** (YOLO, SSD) is highly effective for PPE compliance but limited by occlusion and lighting.

- **Temporal networks** (LSTM, GRU, 3D CNNs) improve fall detection but often require high compute resources.
- **Multi-object tracking** aids proximity detection but is sensitive to camera calibration.
- **Pose estimation** provides ergonomic insights but struggles with camera viewpoint diversity.

The reviewed systems show promising results but often target only **single hazards**, require **high-end GPUs**, or depend on **specialised hardware**.

D. Similar Applications and Existing Solutions.

Although several commercial and research-based systems exist for workplace safety, many exhibit critical limitations.

.1Intenseye

Figure [1] Intenseye Company

is a cloud-based platform focused on monitoring PPE compliance. It offers high detection accuracy and a user-friendly dashboard that simplifies safety management. Nevertheless, the system relies on cloud streaming, making it unsuitable in environments with limited network bandwidth or strict privacy requirements.

2. Protex AI

Figure [2] Protex Company

is another commercial solution that tracks safety incidents such as PPE violations and near-miss events. While it delivers enterprise-grade analytics, it does not support posture or ergonomic assessment, and its high cost can be prohibitive for many companies.

3. SmartVid.io

Figure [3] StarVid.IO Company

is an AI-based safety analytics platform designed for construction sites. It offers robust hazard classification capabilities and can detect unsafe acts effectively, providing safety officers with actionable insights. However, it is a paid service and lacks flexibility for local deployment, which limits its accessibility for smaller organisations or projects in regions with limited resources.

4. YOLO-Based Academic Prototypes.

Figure [4] Ultralytics Company

use object detection models for PPE and hazard detection. Modern versions like **Ultralytics YOLOv5 and YOLOv8** offer fast, accurate real-time detection and edge deployment. However, most academic implementations focus only on detection and lack dashboards, alerts, or multi-hazard integration, limiting practical workplace use.

E. Research Gap

Despite advancements, several key gaps remain:

1. Lack of Unified Multi-Hazard Systems

Most existing systems address **only one hazard**—PPE, falls, or proximity—but not a combined pipeline.

2. Limited Real-Time Performance on Affordable Hardware

High-end models demand powerful GPUs, making them inaccessible for many workplaces.

3. Inadequate Support for Non-Ideal Conditions

Challenges include:

- Low light
- Occlusion
- Non-frontal camera angles
- Crowded industrial scenes

4. Absence of Open, Local Deployment Solutions

Many systems require cloud processing, which is unsuitable for:

- Privacy-sensitive environments
- Poor network stability

- Cost constraints

5. Lack of Integrated Dashboard & Analytics

Research often focuses on detection models but ignores real-world usability elements such as:

- Alert logging
- Incident querying
- Live dashboards
- Worker behavior timelines

VisionSafe 360 directly addresses these gaps.

F. Contribution of VisionSafe 360

VisionSafe 360 provides the following contributions:

1. Multi-Hazard Real-Time Monitoring System

A unified pipeline that detects:

- PPE compliance
- Fall events
- Worker–vehicle proximity risks
- Ergonomic posture deviations

2. Optimized for Existing CCTV Infrastructure

No special sensors or IR cameras required; works on standard RGB streams (RTSP).

3. Efficiency-Focused Deep Learning Models

YOLO-based detection + lightweight temporal networks to enable near real-time operation (>15 FPS on common GPUs).

4. Practical Dashboard for Safety Officers

Including:

- Real-time alerts
- Event logs
- Video evidence
- Hazard analytics

5. Scalable and Modular Architecture

Each module (PPE, fall, tracking, posture) can be improved independently.

Research Gap vs. VisionSafe 360 Contributions

Feature	Manual Monitoring	Commercial AI Platforms	VisionSafe 360 (Pro
Real-Time Detection	No (Reactive)	Yes	Yes
Multi-Hazard Support	High (Human judgment)	Varies (Usually Modular/Paid)	Unified (PPE, Falls,
Cost	High (Man-hours)	High (Licensing/Subscription)	Low (Open Source/
Data Privacy	N/A	Cloud-based (Data leaves site)	On-Premises (Data s
Customisation	Flexible	Low (Proprietary)	High (Modular Arch

Chapter 3

System Analysis

3.1 Problem Statement

Industrial and construction sites are inherently high-risk environments where the failure to detect hazards promptly can lead to severe injuries or fatalities. Traditional monitoring methods, such as manual human supervision and retrospective CCTV review, are reactive by nature. They are limited by human attention span, visual coverage gaps, and the latency between an incident occurring and a response being initiated.

In addition to acute incidents such as falls or collisions, workers are also exposed to long-term **musculoskeletal risks** arising from repetitive, awkward, or sustained postures (e.g., excessive trunk flexion, overreaching, and asymmetric loading), which often remain undocumented and undetected by traditional monitoring practices.

VisionSafe 360 is proposed as an automated, privacy-preserving solution designed to transform existing RTSP CCTV infrastructure into an active safety assistant. By leveraging Edge AI, the system detects four primary categories of safety hazards—**PPE violations**, **fall events**, **unsafe proximity**, and **posture-related ergonomic risks**—in near real-time. The primary objective is to shift industrial safety practices from **reactive investigation** to **proactive prevention**, utilizing localized interfaces tailored for the target operating environment.

3.2 Problem Analysis

A. Constraints on Time and Resources

System Optimization: Real-time operation mandates highly optimized inference pipelines. The target throughput is **15–30 FPS** for optimized models, though actual performance is contingent on model complexity, input resolution, and hardware acceleration.

Hardware Sizing (Prototype Estimates): Based on prototype testing, the approximate resource requirements are estimated at **~2 GB GPU VRAM and ~2.4 GHz CPU per active camera stream**. These figures serve as capacity planning estimates; the system is designed to scale dynamically (e.g., via frame skipping or clustering) based on available compute power.

Data Reliability: Variability in industrial lighting, occlusion, and camera angles necessitates robust dataset curation and per-site threshold calibration to minimize false positives, both for acute hazard detection and for reliable estimation of **posture-related musculoskeletal risks** across diverse operating conditions.

B. Safety and Operational Barriers

Surveillance Fatigue: Human operators cannot maintain continuous, error-free attention across multiple video feeds for extended periods.

Reactive Latency: Post-incident video review is valuable for forensics but fails to prevent immediate harm.

Documentation Burden: Manual incident reporting is time-consuming and prone to human error. Automation is required to produce accurate, audit-ready logs covering both acute events (falls, PPE, proximity) and prolonged exposure to **ergonomically harmful postures**.

C. Current Solutions and Limitations

Manual Supervision: Effective only for small, contained areas; lacks scalability.

Cloud-Based AI: Often requires streaming raw video off-site, raising significant concerns regarding **bandwidth costs** and **data privacy**. Latency in cloud uploading can also delay critical alerts.

D. Technological Challenges

Real-Time Processing: The system must perform unified hazard detection—covering object detection, tracking, posture estimation, and rule evaluation (e.g., proximity logic)—concurrently within a sub-second latency budget.

Privacy Compliance: To comply with data protection standards, **on-edge anonymization (face blurring)** is mandatory before any visual data is persisted or transmitted. This also applies to posture analysis: only anonymized skeletal keypoints and ergonomic scores are propagated to the backend, not raw images.

3.3 Project Feasibility Study

A. Technical Feasibility

The system is built upon a proven open-source stack: **Python**, **OpenCV** (video ingestion), **Ultralytics YOLOv8** (object detection), and **FastAPI** (backend services). The architecture utilizes **Edge Computing** for critical inference, ensuring feasibility even in bandwidth-constrained environments.

In addition to PPE, fall, and proximity detection, the Edge AI stack is extended with a **Posture & Ergonomic Analysis sub-module**. This module relies on skeletal keypoint extraction (e.g., from a lightweight pose estimation network) and rule-based ergonomic evaluation to approximate posture-related risk levels in real-time.

Within the Edge AI Engine, all four hazard categories—PPE violations, falls, unsafe proximity, and posture-related ergonomic risks—are evaluated in a unified pipeline. The engine not only detects the presence of hazards but also classifies their **severity** (e.g., high, medium, low), enabling a differentiated alert policy that reserves emergency alerts for acute hazards while treating posture hazards as low-severity, preventive signals. As with other AI components, all analysis runs locally on the Edge node, ensuring that raw video remains on-premises and only anonymized metadata is transmitted to the backend for long-term analysis.

B. Economic Feasibility

VisionSafe 360 leverages existing CCTV hardware, significantly reducing Capital Expenditure (CAPEX). The Return on Investment (ROI) is driven by the reduction in workplace accidents, lower insurance liabilities, and decreased administrative overhead for HSE teams, as well as by reducing long-term **musculoskeletal exposure** through preventive posture monitoring and ergonomic risk analysis.

C. Legal Feasibility

The design explicitly addresses privacy laws by performing local processing. No raw video streams are recorded or uploaded; only anonymized metadata and encrypted snapshots are retained, adhering to strict data privacy regulations for both acute incident data and long-term ergonomic posture records.

D. Operational Feasibility

The platform is designed to complement, not replace, existing HSE workflows. The localized Dashboard and Mobile App require minimal training, ensuring high adoptability by site supervisors and safety engineers, both for managing real-time incidents and for reviewing **posture-related ergonomic risk trends**.

3.4 Project Time Scheduling

The project follows a modular development lifecycle designed to allow parallel development of independent components (e.g., frontend and AI model training). The schedule is divided into six main phases:

Requirements Analysis & Dataset Preparation: Defining scope and collecting diverse industrial datasets, including scenarios for PPE, falls, unsafe proximity, and representative working postures with ergonomic implications.

Prototype Edge Inference Development: Building the core detection engine.

Backend Services & Database Design: Implementing the API gateway and storage schema.

Real-time Alerting & Mobile Integration: Connecting the Edge engine to user devices.

Dashboard Analytics & Reporting: Developing the visualization layer, including safety indicators and **ergonomic risk reports**.

Integration Testing & Validation: Full system stress testing.

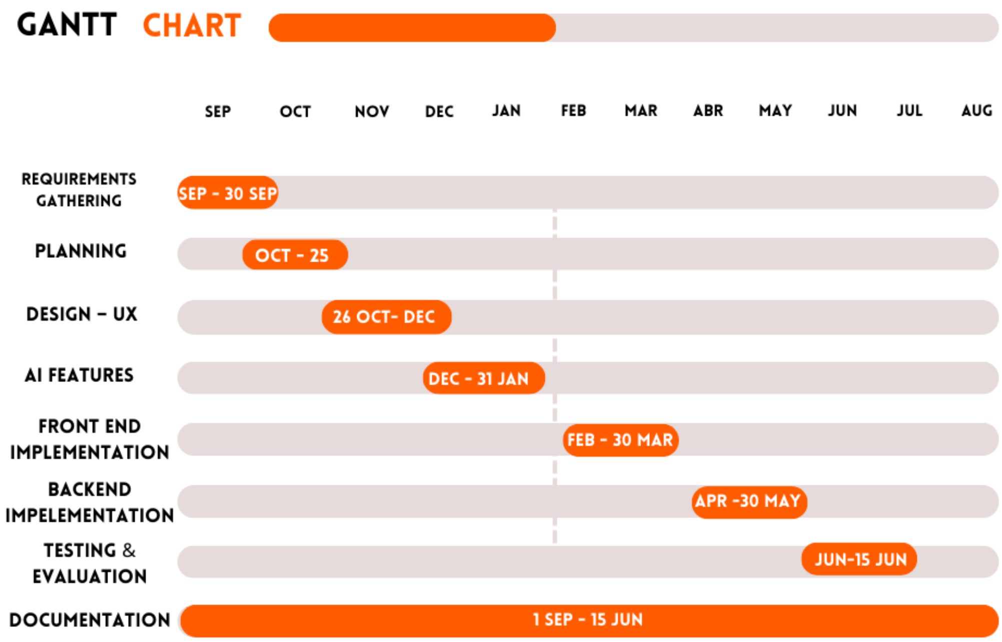


Figure [5] Gantt Chart of Project Schedule

3.5 Functional Requirements

User Authentication & RBAC:

Secure login supporting distinct roles: **Admin**, **HSE Engineer**, **Site Supervisor**, and **Data Analyst**.

Admins possess full privileges for system configuration and user management.

Camera Management:

Admins can add/edit RTSP streams via the Dashboard.

System validates stream connectivity immediately upon entry.

Metadata includes: camera_id, RTSP_URL, ROI_polygon, and location_tag.

Edge AI Engine (Internal Component):

Autonomous execution of unified hazard inference pipelines for PPE detection, fall detection, unsafe proximity detection, and posture-related ergonomic risk analysis.

Face Blurring: Applied automatically to all detected persons prior to snapshot generation.

Hazard Severity Classification: For each detected hazard (PPE, fall, proximity, posture), the engine assigns an appropriate severity level (e.g., high, medium, low) based on confidence, duration, and hazard type. These severity levels are used by the alerting logic to differentiate between emergency incidents and low-severity ergonomic records.

Real-Time Alerting:

Local (Emergency Incidents): Immediate activation of on-site sirens/lights (latency <1s) **only** for acute, high-severity safety hazards, namely:

Confirmed fall events.

PPE violations (e.g., missing helmet or vest) above configured thresholds.

Unsafe proximity between workers and hazardous zones or moving equipment.

Remote (Emergency Incidents): Push notifications (FCM) to Supervisors and WebSocket updates to the HQ Dashboard are generated for the same set of acute, high-severity hazard types (falls, PPE violations, unsafe proximity).

Non-Emergency Ergonomic Alerts: Posture-related ergonomic hazards, typically classified as low or medium severity, are logged as **low-severity ergonomic events** and surfaced in dashboards and reports. They **do not** trigger sirens or high-priority mobile alerts, thereby avoiding alert fatigue while still providing actionable insights for preventive ergonomics.

Event Management & Lifecycle:

Structured logging of incidents with timestamp, confidence score, computed severity, and snapshot.

Closed-Loop Workflow: Supervisors must ”Acknowledge” alerts and record ”Action Taken” to resolve an incident.

Health Monitoring:

Continuous tracking of camera status (Online/Offline), FPS, and connectivity.

Automated reconnection logic with exponential backoff.

Privacy & Retention:

Strict policy: **No raw video storage**.

Configurable retention periods for snapshots and metadata, including both incident and ergonomic posture records.

Posture & Ergonomic Risk Assessment:

The system shall continuously estimate worker body posture using skeletal keypoints extracted from video frames (e.g., shoulders, hips, knees, trunk orientation).

An internal ergonomic scoring mechanism (inspired by standard frameworks such as **RULA/REBA**) shall classify posture snapshots into low, medium, or high musculoskeletal risk levels.

Non-emergency **ergonomic alerts** shall be generated for prolonged high-risk postures (e.g., sustained trunk flexion beyond a configurable duration). These are logged as **low-severity events** and do not activate emergency sirens or high-priority mobile alerts.

Aggregated posture risk data (per worker zone or per camera) shall be stored as **ergonomic records** and made available to HSE Engineers for trend analysis and preventive intervention planning, helping identify high-risk tasks and areas over time without causing alert fatigue.

3.6 Non-Functional Requirements

Performance:

Target Inference Rate: **15–30 FPS** (hardware dependent).

End-to-End Latency (Detection to Mobile Alert): **2–3 seconds**.

Scalability: Horizontal scalability achieved by adding Edge nodes; vertical scalability via model optimization (e.g., quantization).

Reliability: The system must recover automatically from network interruptions and camera signal loss, for both real-time incident detection and continuous posture data collection.

Security:

Data in Transit: TLS/SSL encryption for API and WebSocket connections.

Data at Rest: Encrypted storage for snapshots; hashed passwords.

3.7 System Actors and Roles

Actor	Responsibilities
System Administrator	System config, user management, camera setup, health monitoring.
HSE Engineer	Dashboard monitoring, incident investigation, compliance auditing, ergonomic tr
Site Supervisor	Receives mobile alerts, performs on-site intervention, logs corrective actions.
Data Analyst	Reviews long-term trends, generates safety KPIs and reports.
AI Engine	Internal System Actor. Performs unified hazard detection (PPE, falls, proximity,

3.8 Use Case Modeling

Use Case Diagram

The Use Case diagram provides a high-level view of the system's functional scope and the interactions between actors and use cases. It clearly distinguishes between human actors (Admin, HSE, Supervisor) who interact with the UI, and the **AI Engine**, which is depicted as a system actor responsible for backend automation. A unified hazard detection use case covers all four hazard categories (PPE violations, falls, unsafe proximity, and posture-related ergonomic risks), while the alerting and reporting interfaces distinguish between emergency incident handling and preventive ergonomic analysis.

While the **AI Engine** executes hazard detection and posture analysis in a unified pipeline on the Edge node, only acute, high-severity hazards (falls, PPE violations, unsafe proximity) enter the emergency alerting workflow; posture-related deviations, typically of lower severity, are captured as ergonomic records and exposed through analytic views rather than immediate alarms.

Figure 3-2: Use Case Diagram

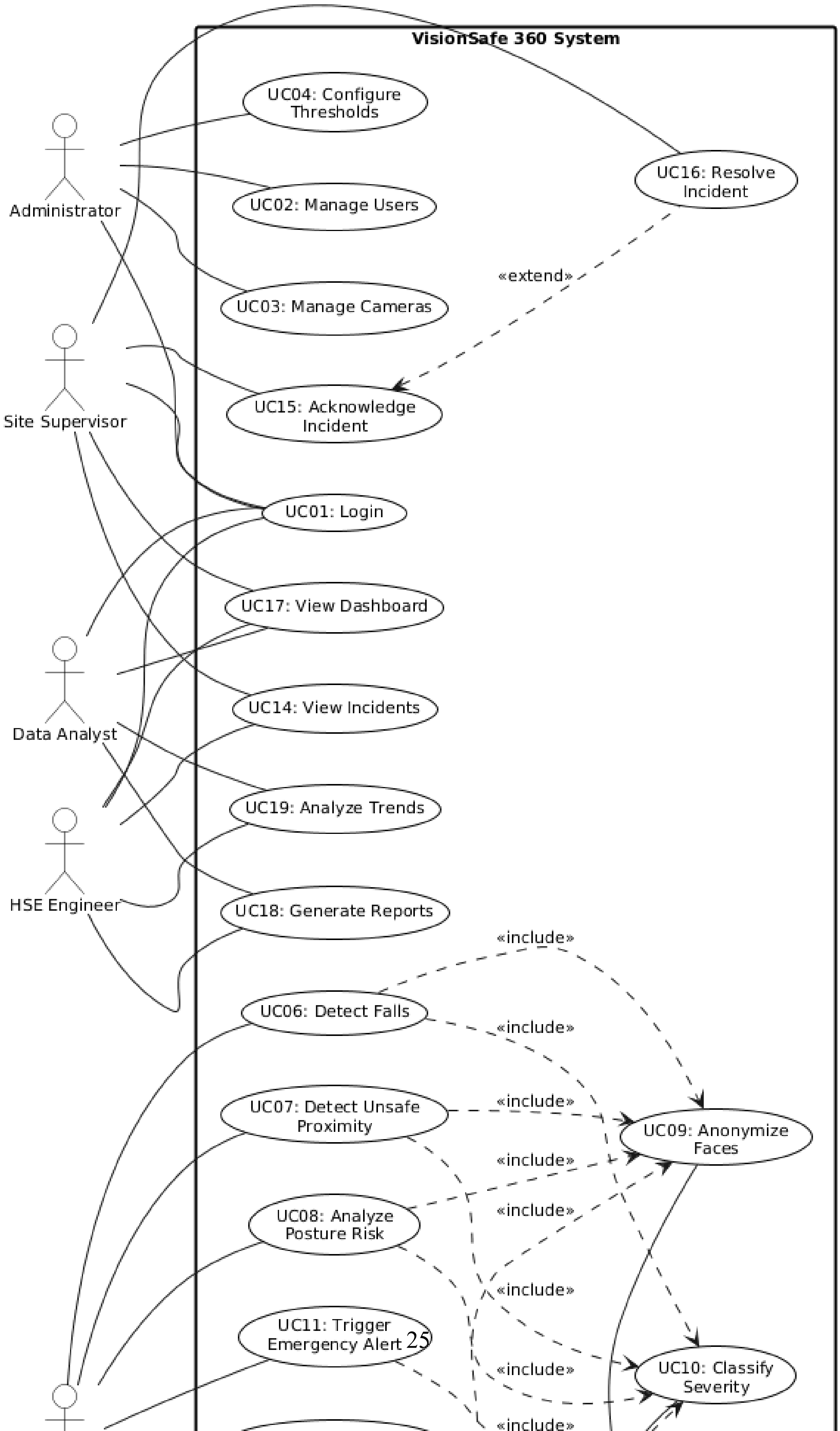


Figure [6] - Key Use Case Scenarios

Hazard Detection: The Edge AI detects a ”Fall”. The system validates confidence, classifies severity, checks the cooldown timer, triggers the local siren, blurs faces in the frame, and uploads the event to the Backend as an emergency incident.

Incident Resolution: A Supervisor receives a notification, opens the app to view the snapshot, attends to the worker, and marks the incident as ”Resolved” with notes.

Camera Onboarding: The Admin inputs an RTSP URL. The system attempts a connection handshake; if successful, the camera is added to the active monitoring pool.

Posture Trend Analysis: The AI Engine periodically evaluates worker posture using skeletal keypoints and assigns ergonomic risk scores (e.g., low/medium/high). These low-severity posture records are aggregated by camera or work zone and later reviewed by HSE Engineers through the dashboard to identify tasks or areas associated with sustained high ergonomic risk.

3.9 Activity Diagram

This diagram details the sequential logic flow of the core detection loop, specifically focusing on how the system handles continuous video frames. It illustrates the critical decision-making process: acquiring a frame, running unified hazard inference (PPE, falls, proximity, posture), validating the detections against confidence thresholds, and then classifying the **severity** of each hazard. This mechanism is vital to prevent ”alert fatigue” by ensuring that only genuinely high-severity, acute hazards (e.g., a confirmed fall with high confidence) can trigger immediate emergency alerts.

In parallel, or rather as part of the unified hazard analysis, posture-related ergonomic hazards are evaluated and assigned typically low or medium severity levels. These posture hazards generate **non-emergency, low-severity posture records** for long-term musculoskeletal risk assessment. These posture-related records do not trigger real-time sirens or high-priority alerts but instead feed ergonomic dashboards and reports for HSE teams.

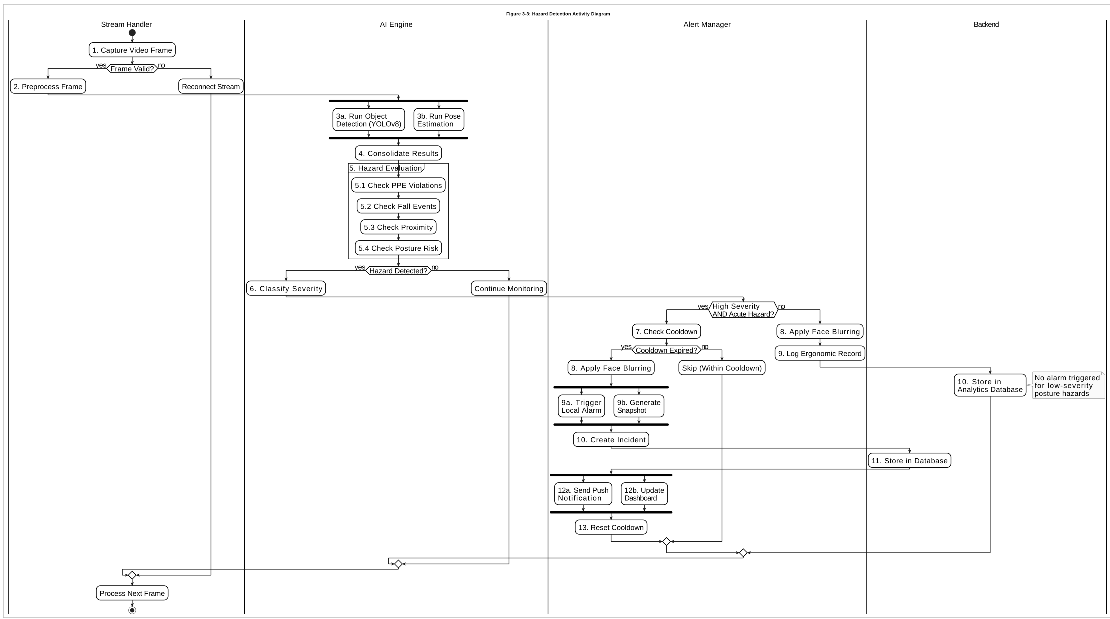


Figure [7] – Activity Diagram

Flow Summary:

Capture Frame → Unified Inference (PPE/Fall/Proximity/Posture) → Hazard Evaluation → Severity Classification →

If High/Critical & Acute Hazard (PPE/Fall/Proximity):

Cooldown Check → Trigger Alarm → Anonymize & Save → Notify Backend (Incidents)

Else (Low/Medium Severity, including most Posture cases):

Log Ergonomic / Low-Severity Record → Notify Backend (Analytics/Ergonomics).

3.10 Data Flow Diagram (DFD)

Level 0 (Context Diagram)

The Context Diagram (Level 0) defines the system boundary of VisionSafe 360. It illustrates the external entities that interact with the system: **Cameras** (Input source), **Site Supervisors/HSE** (Notification recipients and ergonomic data consumers), and **Admins** (Configuration). The central process represents the entire VisionSafe system, abstracting internal complexities to focus on input/output data streams, including both real-time emergency incident alerts and longer-term ergonomic posture records.

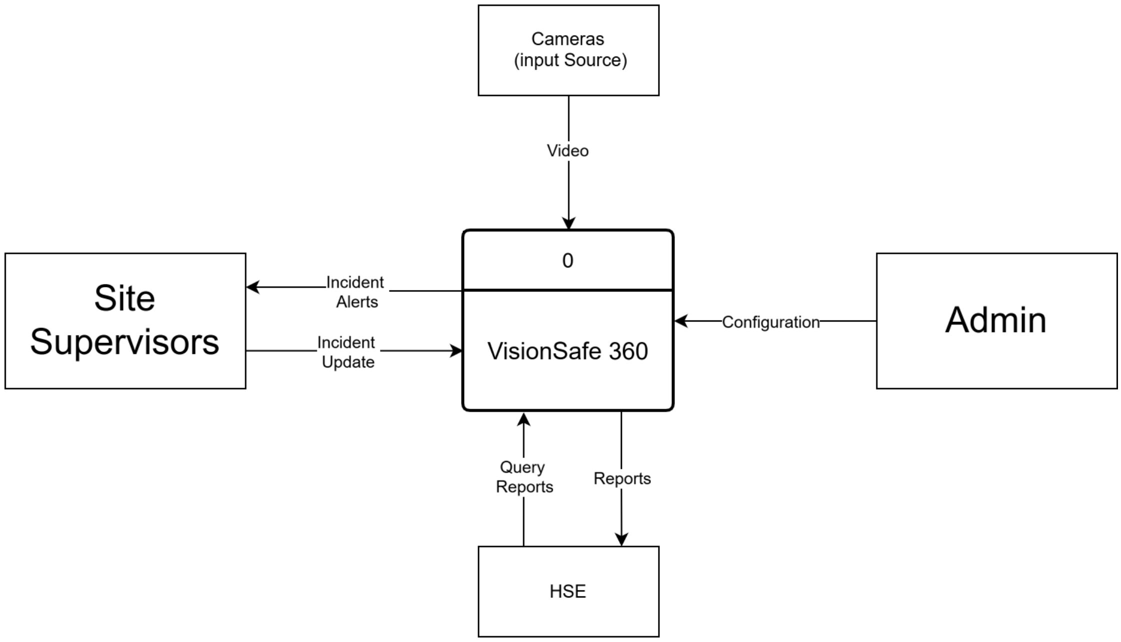


Figure [8] – Context Diagram

Level 1 Diagram

The Level 1 DFD decomposes the system into its primary subsystems to show detailed data routing:

Ingestion & Processing: RTSP streams enter the Edge module where unified hazard inference (PPE, falls, proximity, posture) and face blurring occur. Hazard evaluation and severity classification are applied to all hazard categories.

Transmission: Only metadata and encrypted snapshots are passed to the **API Gateway**. This includes both hazard metadata for acute, high-severity incidents (e.g., falls, PPE violations, unsafe proximity) and posture-related ergonomic scores and durations (typically logged as low-severity records for trend analysis).

Storage: Structured data is persisted in the SQL Database, with separate fields or records representing emergency incidents (e.g., falls, PPE violations) and non-emergency ergonomic observations (e.g., high-risk posture episodes).

Presentation: The Backend serves data to the Web Dashboard and pushes notifications to Mobile Clients. HSE Engineers can review both real-time emergency alerts (for acute hazards) and aggregated low-severity ergonomic posture trends to inform preventive interventions without overwhelming supervisors with immediate alarms.

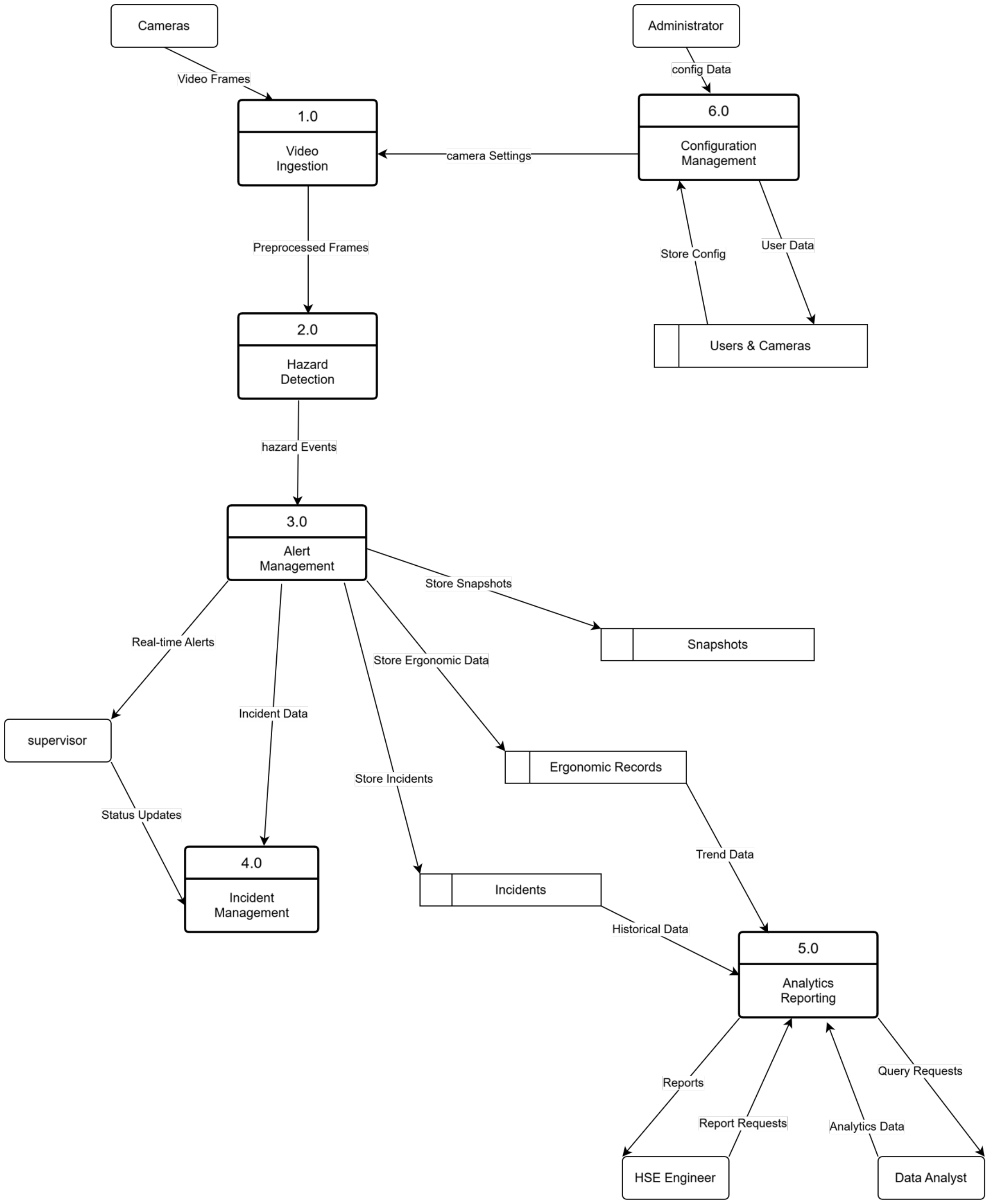


Figure [9] – Data Flow Diagram (Level 1)

3.11 Entity Relationship Diagram (ERD)

The ERD visualizes the database schema designed to support the system's data persistence requirements. It employs a normalized relational structure to ensure data integrity. The central entity is the **Incident**, which links **Cameras** (where it happened), **Hazard Types** (what happened), and **Users** (who resolved it). This structure supports complex queries for analytics, such as "Total PPE violations per camera per week". Posture-related ergonomic risks are represented as hazard types of lower severity (e.g., POSTURE_RISK) and can be queried independently or alongside other hazards to understand both acute and chronic risk patterns.

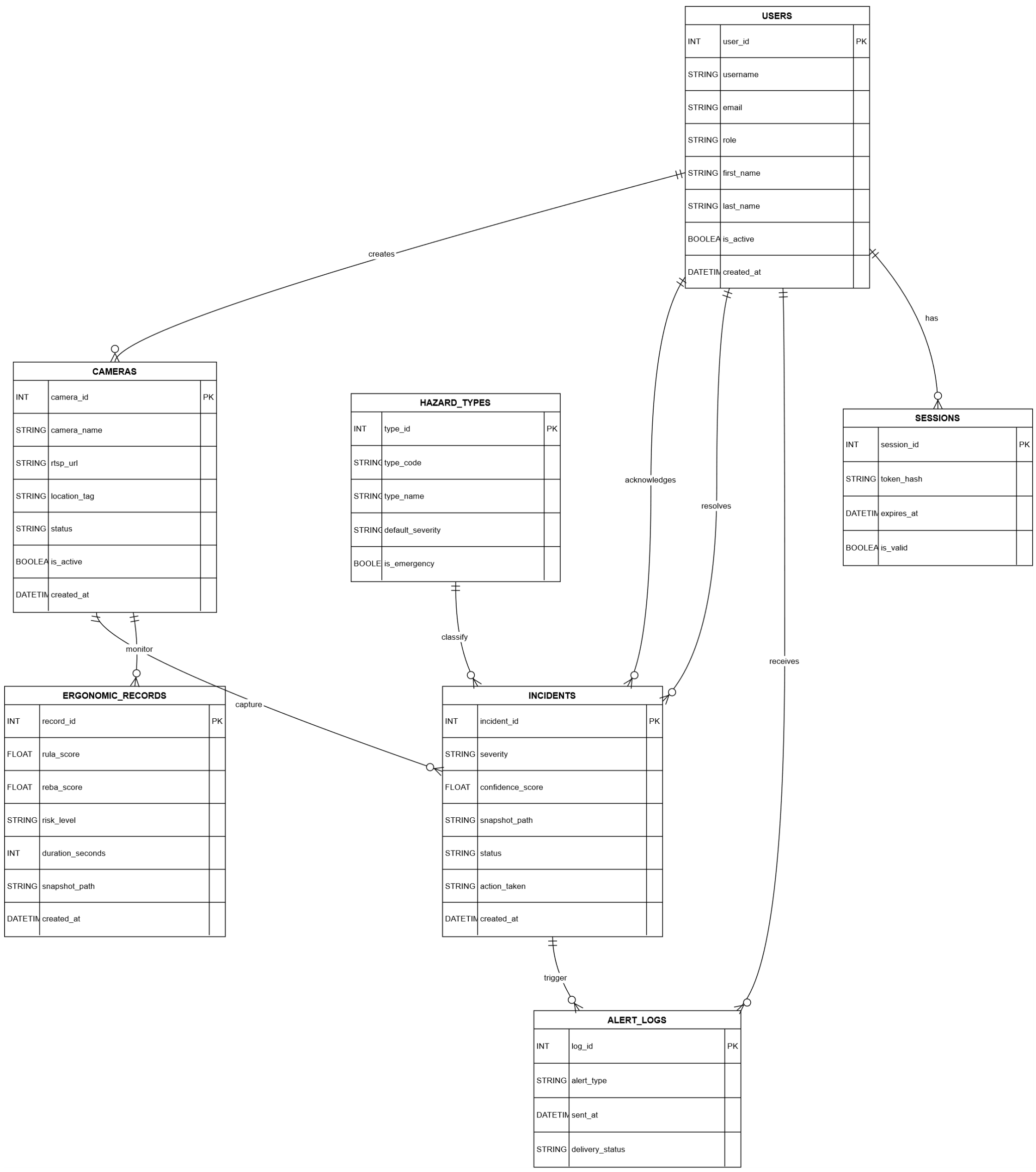


Figure [10] – Entity Relationship Diagram

Key Entities:

Users: Stores authentication credentials, roles, and FCM tokens for mobile pushes.

Cameras: Contains configuration data including RTSP URLs and JSON-defined Region of Interest (ROI) polygons.

Incidents: The core transactional table storing event timestamps, confidence scores, severity, snapshot paths, and resolution status. This includes safety-critical events (e.g., falls) and may also reference posture-related events classified as low-severity when needed for ergonomic analysis.

HazardTypes: A lookup table defining detectable classes (e.g., ”No Helmet”, ”Fall”,

”Proximity”, ”Posture Risk”) and their associated default severitylevels.

3.12 Class Diagram

The Class Diagram represents the static structure of the software implementation, adhering to Object-Oriented Design (OOD) principles. It highlights the modular separation of concerns:

StreamHandler: Responsible solely for video I/O, frame buffering, and connection health.

InferenceEngine: Wraps the YOLO model (and any auxiliary pose estimation models), handling pre-processing and post-processing of tensors.

HazardAnalyzer: Contains the business logic for unified hazard evaluation (PPE, falls, proximity, posture). It generates hazard events enriched with type and severity information.

PostureAnalyzer: A dedicated analytic component that consumes frames and/or pose data to extract skeletal keypoints, compute ergonomic scores (e.g., RULA/REBA-inspired), and provide posture-related hazard candidates and risk levels to the HazardAnalyzer.

Alerting/Notification Logic (e.g., AlertManager + NotificationService): Applies an alert routing policy based on hazard severity and type:

High/Critical severity for acute hazards (falls, PPE, proximity) → emergency sirens and incident creation.

Low/Medium severity, especially posture-related ergonomic hazards → ergonomic records and analytic logging only (no sirens).

NotificationService: An interface for triggering external actions (Sirens, API calls) without coupling the core logic to specific output hardware.

The PostureAnalyzer and HazardAnalyzer jointly implement a unified hazard detection pipeline, while the alerting logic enforces different behaviors for emergency incidents versus low-severity ergonomic records.

Figure 3-7: Class Diagram

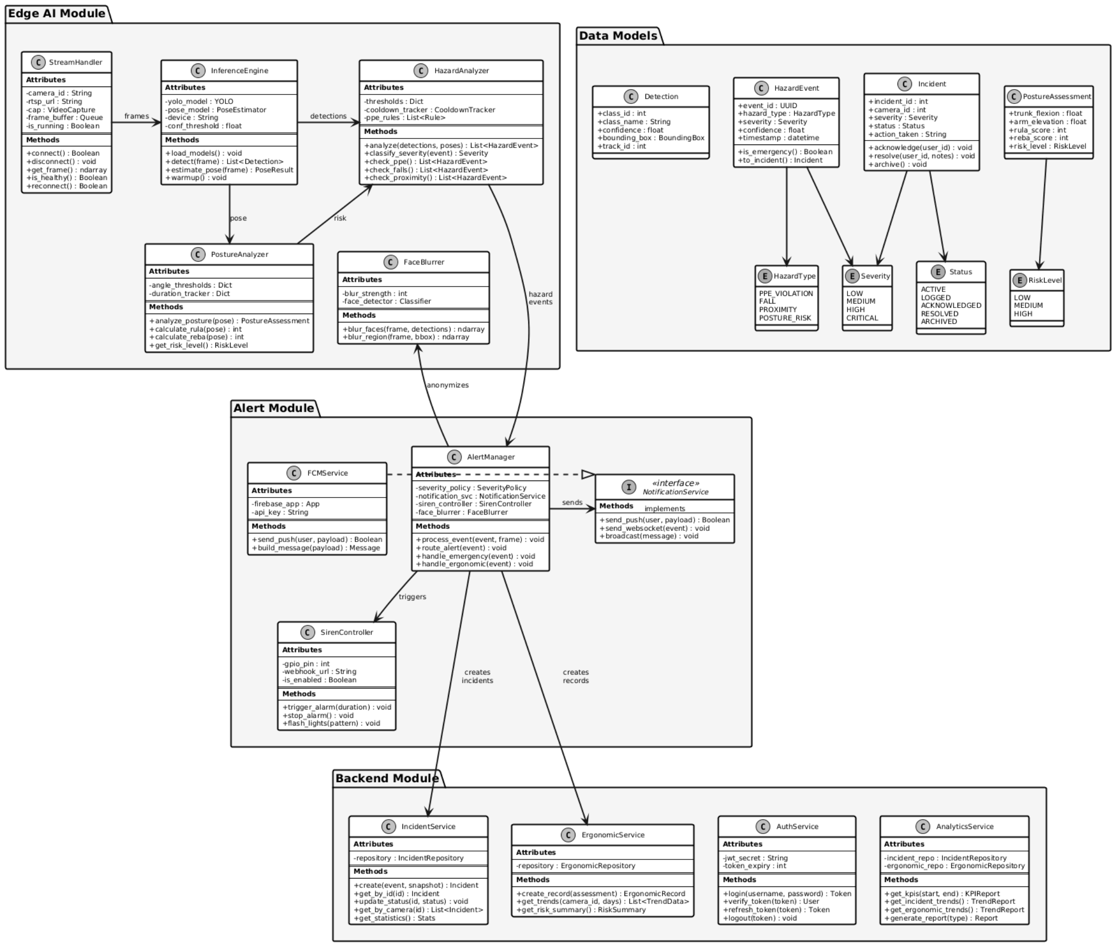


Figure [11] – Class Diagram

3.13 Sequence Diagram

The Sequence Diagram provides a dynamic view of the system, capturing the chronological order of interactions during a ”Hazard Detection” scenario. It traces the lifecycle of a single video frame: from capture by the **StreamHandler**, through unified hazard analysis by the **InferenceEngine**, **PostureAnalyzer**, and **HazardAnalyzer**, to the final alerts triggered (or not triggered) by the alerting logic and received by the **Mobile Client**.

In this workflow, a unified hazard event is created, containing both the hazard type (PPE, fall, proximity, posture) and its severity (e.g., high/medium/low). The alerting logic then decides whether this event becomes an emergency incident (with sirens and mobile alerts) or a low-severity ergonomic record suitable for trend analysis. This diagram is critical for understanding the latency and dependencies between the Edge and Backend components, as well as the separation between emergency incident handling (high-severity PPE/fall/proximity) and continuous, preventive ergonomic analysis (typically low-severity posture hazards) that avoids contributing to alert fatigue.

Figure 3-8: Hazard Detection Sequence Diagram

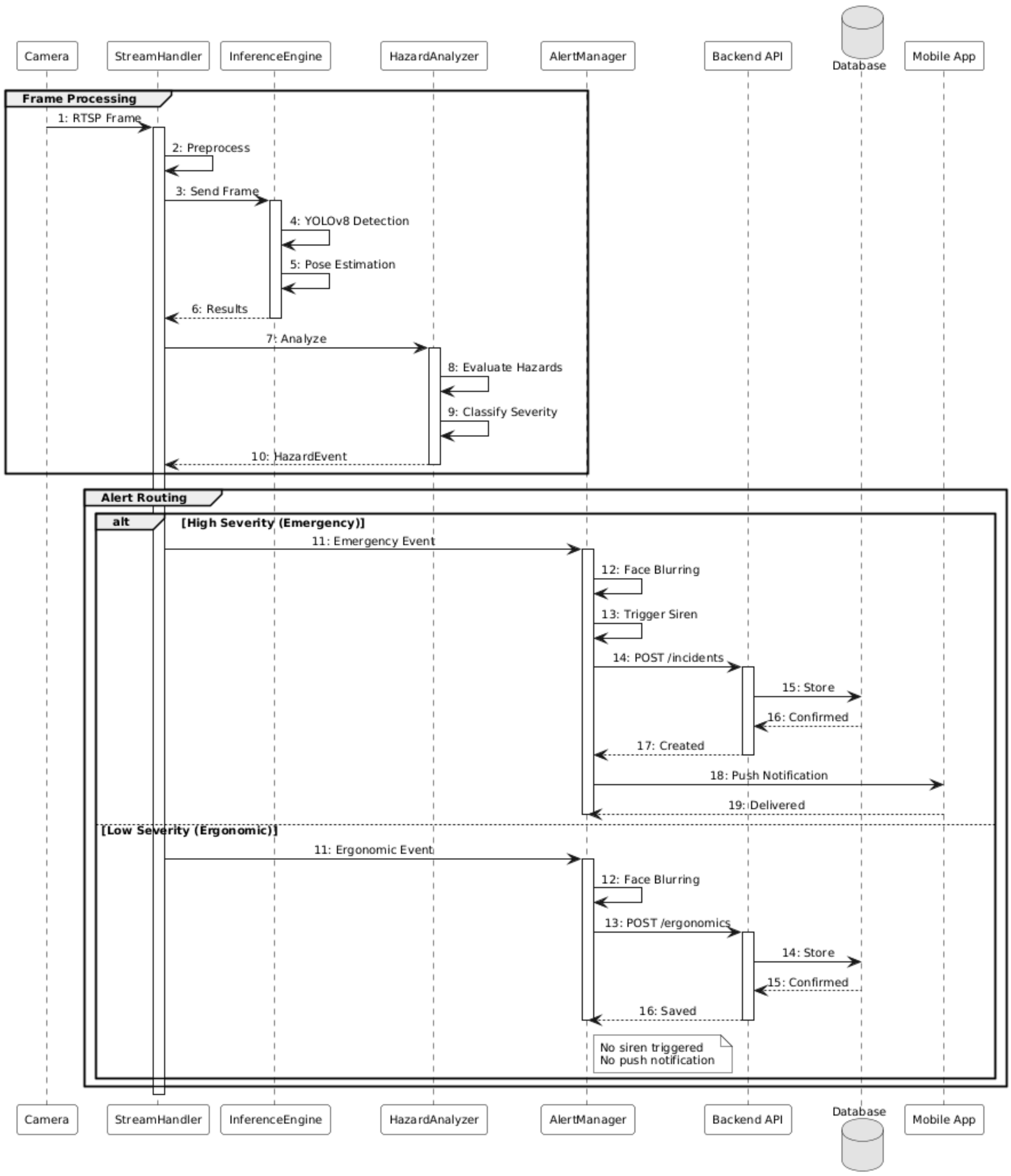


Figure [12] – Sequence Diagram

Sequence Steps:

StreamHandler provides a frame to **InferenceEngine**.

InferenceEngine returns detection bounding boxes and pose-related keypoints to **HazardAnalyzer** and **PostureAnalyzer**.

PostureAnalyzer evaluates skeletal keypoints, computes ergonomic scores, and provides posture-related hazard candidates to **HazardAnalyzer**.

HazardAnalyzer consolidates PPE, fall, proximity, and posture-related inputs, confirms validity (rules + thresholds), and produces hazard events enriched with **hazard type and severity**.

The alerting logic (e.g., AlertManager + NotificationService) inspects each hazard event:

For high-severity acute hazards (e.g., falls, PPE, proximity), it triggers local hardware (sirens/lights) and posts an incident to the **Backend API**.

For low-severity posture-related ergonomic hazards, it bypasses sirens and posts an ergonomic record to the **Backend API** asynchronously.

Backend persists both emergency incidents and ergonomic records and pushes a real-time update to the **Mobile Client** (for incidents) and posture dashboards (for HSE Engineers).

3.14 Incident Lifecycle & State Transition

This State Transition Diagram models the lifecycle of an Incident object, ensuring accountability and workflow enforcement. It defines how an incident moves from being a raw detection to a closed case. This prevents incidents from being ”lost” and mandates human intervention for closure.

Figure 3-9: Incident Lifecycle State Diagram

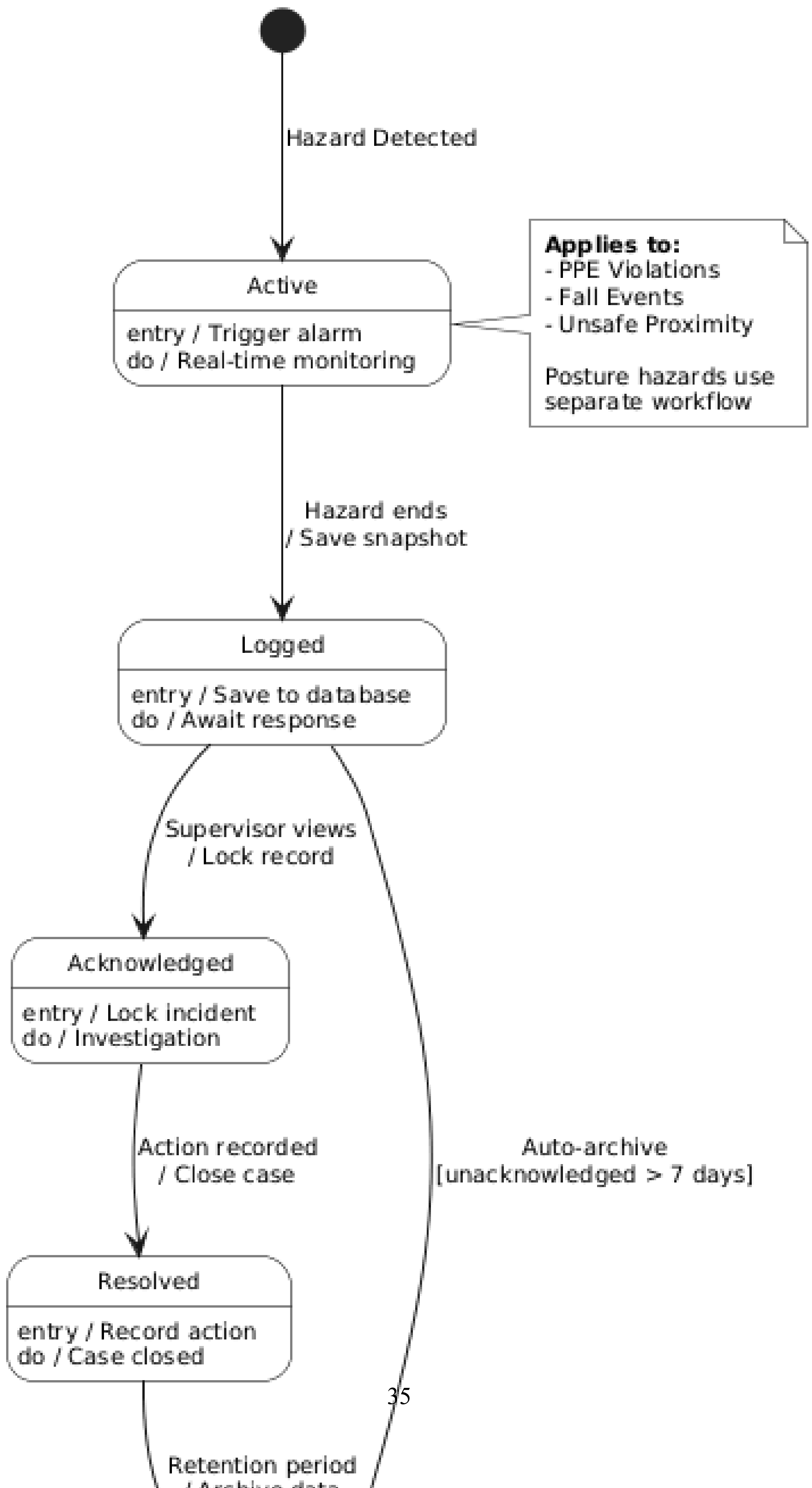


Figure [13] – State Diagram

Active: The hazard is currently detected; alerts are firing.

Logged: The hazard has ceased or stabilized; the event is saved in the DB but requires attention.

Acknowledged: A Supervisor has viewed the notification (locking the incident).

Resolved: Corrective action has been logged, and the case is closed.

Archived: Old data is moved to cold storage for long-term retention.

The Incident lifecycle focuses on acute, high-severity, safety-critical events (such as PPE violations, falls, and unsafe proximity). Posture-related ergonomic observations, typically classified as low or medium severity, are treated as **non-critical records** and do not enter the emergency incident lifecycle. Instead, they are stored as ergonomic events associated with cameras or work zones, supporting long-term analysis and preventive ergonomics rather than immediate alarm escalation.

3.15 Component Diagram

The Component Diagram illustrates the high-level modular decomposition of VisionSafe 360 and clarifies the responsibilities and interfaces between subsystems across the Edge and Backend environments. It emphasizes the system's design principle of keeping safety-critical inference and privacy enforcement on-premise (Edge), while delegating persistence, analytics, and multi-client distribution to the Backend.

At the Edge Node, the Stream Handler ingests RTSP camera streams and provides frames to the AI Engine for unified hazard inference. The Posture Analyzer operates alongside the core inference pipeline to estimate skeletal keypoints and compute ergonomic risk indicators, which are then consolidated within the hazard evaluation flow. Prior to any persistence or transmission, the Face Blurring component enforces privacy by anonymizing detected persons. An Alert Controller orchestrates event routing, ensuring that high-severity acute hazards trigger local alarms (via GPIO) while posture-related ergonomic risks are typically logged as low-severity records for trend analysis.

On the Backend Server, an API Gateway exposes secured endpoints and routes requests to specialized services, including Authentication, Incident Management, Ergonomic Records, and Analytics. Data persistence is handled through a dedicated Data Layer, where structured records are stored in PostgreSQL, frequently accessed data is optimized using Redis cache, and encrypted snapshots are stored in file storage. External services such as Firebase FCM are integrated to deliver push notifications to client applications, including the Mobile App and Web Dashboard.

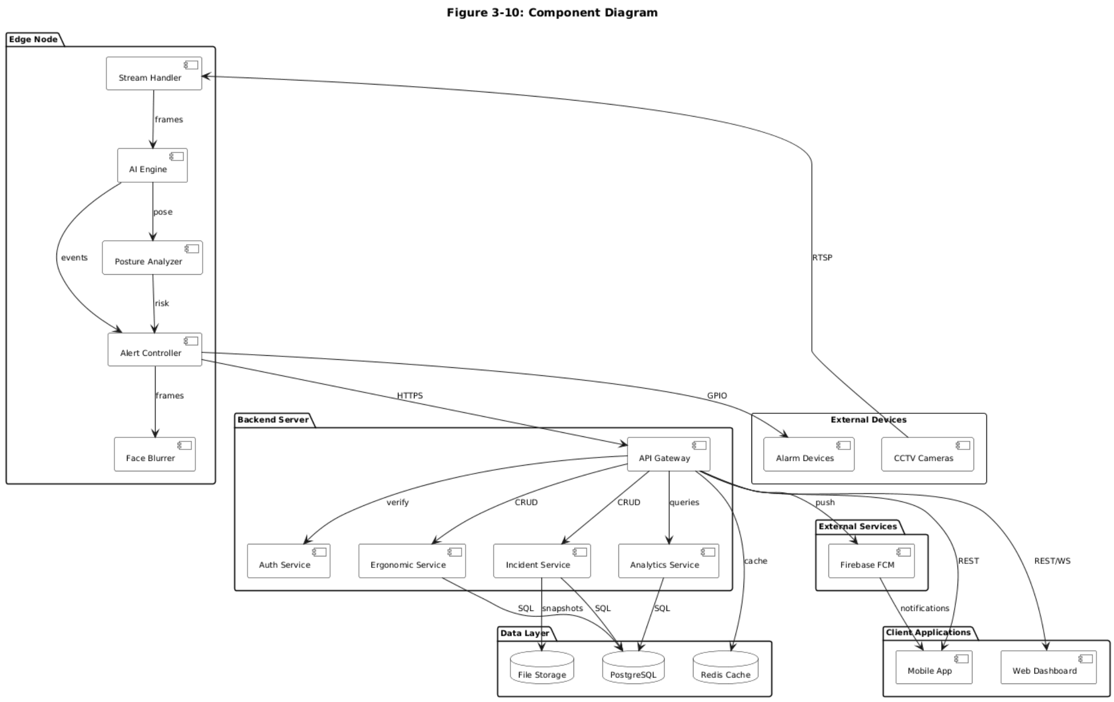


Figure [14] – Component Diagram

3.16 Deployment Diagram

The Deployment Diagram presents the physical runtime topology of VisionSafe 360 across an industrial site and cloud infrastructure. It highlights how the system integrates with existing CCTV installations while ensuring low latency for emergency response and preserving privacy by keeping raw video on-site.

Within the Industrial Site, multiple IP cameras stream video over RTSP to the Edge Server (e.g., NVIDIA Jetson). The Edge node hosts the core runtime modules—stream handling, inference execution, and alert management—along with the primary AI models (YOLOv8 for detection and a lightweight pose model for posture estimation). For immediate on-site response, the Edge Server interfaces with the Alarm System (e.g., siren and strobe light) through GPIO, enabling sub-second activation for confirmed high-severity hazards.

The Cloud Infrastructure hosts the application layer responsible for orchestration and user-facing services, including the FastAPI Backend and a WebSocket server for real-time dashboard updates. Persistent storage is provided through PostgreSQL for structured incident/ergonomic records, Redis for caching and performance optimization, and dedicated file storage for encrypted snapshots. Firebase FCM is used as an external notification layer for mobile push delivery. Client devices (desktop web browser and iOS/Android mobile app) communicate with the backend using secure channels (HTTPS/WSS), receiving emergency incident notifications and accessing analytics views, including aggregated ergonomic posture risk trends.

This deployment layout supports scalability (by adding Edge nodes per site/camera cluster), ensures operational resilience under limited connectivity, and enforces privacy compliance by restricting raw video processing and anonymization to the on-premise Edge environment.

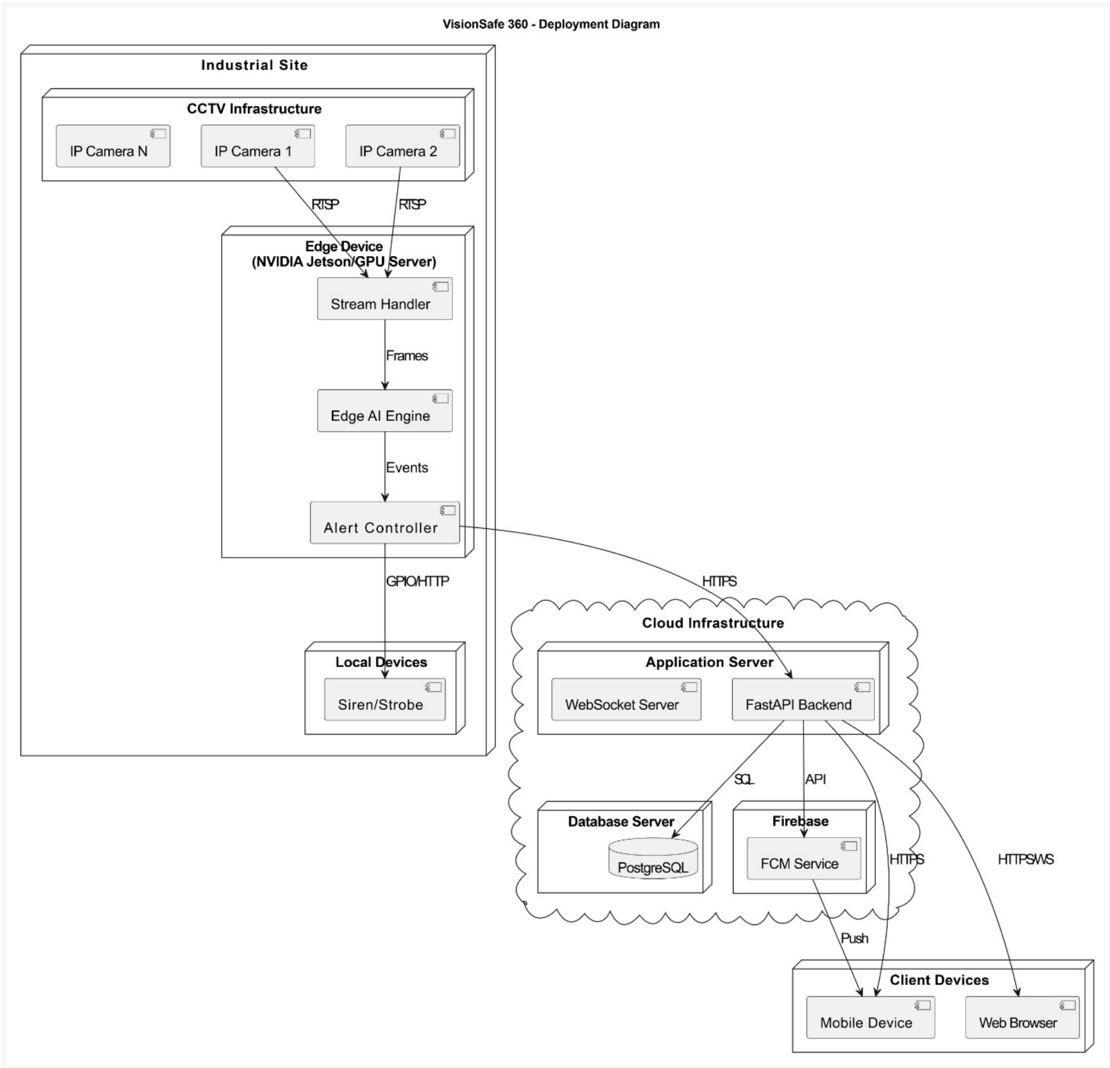


Figure [15] – Deployment Diagram

3.17 API Specification Summary

The communication between the Edge module, Dashboard, and Mobile App is mediated by a RESTful API. The following summary outlines the primary endpoints, all secured via JWT authentication:

POST /auth/login: Authenticates users and returns access tokens.

GET /cameras/health: Returns real-time status (Online/Offline/FPS) of all streams.

POST /incidents: Internal endpoint for the Edge Engine to upload new emergency incidents (generally high-severity PPE, fall, and proximity hazards).

PATCH /incidents/{id}: Used by Supervisors to update incident status and add "Action Taken" notes.

WS /events: A WebSocket endpoint providing a live feed of emergency incident events to the dashboard.

(Optionally) POST /ergonomics: Internal endpoint for uploading **non-emergency, low-severity posture records and ergonomic risk scores** for trend analysis, feeding dashboards and reports rather than immediate emergency alerting.

3.18 Chapter Summary

This chapter provided a comprehensive analysis of VisionSafe 360. It defined the problem scope, analyzed operational constraints, and detailed the functional and non-functional requirements necessary for a robust safety solution. The system architecture-prioritizing **Edge computing** for latency and privacy—was modeled through standard UML diagrams (Use Case, Activity, Class, Sequence) and Data Flow Diagrams.

The design incorporates a **unified, AI-driven hazard detection pipeline** covering four hazard categories: **PPE detection**, **fall detection**, **unsafe proximity monitoring**, and **Posture-related Ergonomic Risk Analysis**. A severity-aware alerting policy clearly distinguishes between emergency incident handling—reserved for high-severity, acute hazards—and continuous, preventive ergonomic risk monitoring for posture-related hazards. Within this architecture, acute hazards (PPE violations, falls, unsafe proximity) drive the real-time emergency alerting workflow, whereas posture-related ergonomic deviations are continuously monitored and recorded as low-severity events for long-term trend analysis and preventive ergonomics, thereby supporting HSE teams without introducing additional alert fatigue. These design specifications lay the groundwork for **Chapter 4**, which will detail the implementation of the inference pipelines, backend services, and interface integration.

Chapter 4: Project Design

Chapter 4

Project Design

4.1 Overview of the Design

VisionSafe 360 is designed as a user-centered safety monitoring ecosystem for industrial and construction environments. The project focuses on converting raw safety observations into actionable intelligence to reduce incident response time and improve long-term preventive safety practices.

The primary objective is to manage four critical workplace risks: PPE compliance, fall events, unsafe proximity, and ergonomic hazards. The design distinguishes between **Emergency Response** (immediate action) and **Preventive Analytics** (long-term trends), ensuring that safety-critical events are surfaced with urgency while minimizing alert fatigue for long-term data. The design ensures a seamless integration between high-level oversight and rapid field-level intervention.

4.2 System Architecture

Overall Architecture

The system is structured as a modular, cross-platform architecture designed to support two distinct interaction environments:

Control Room Environment (Web Dashboard): Optimized for continuous situational awareness, detailed incident investigation, and administrative oversight.

On-Site Response Environment (Mobile App): Tailored for rapid alert reception, immediate acknowledgement, and field-level resolution logging.

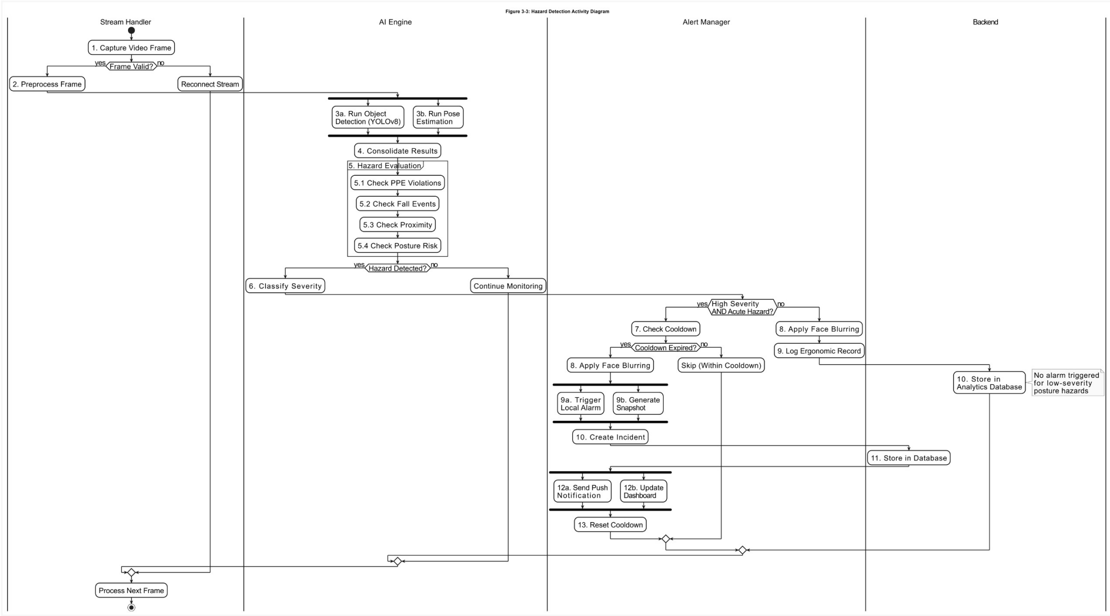


Figure [16]: System Architecture

Component Roles and Interactions

a. **Data Acquisition & Processing:** Integrates camera feeds and AI models to detect safety breaches.

b. **Event Presentation Layer:** Translates system detections into human-readable summaries (type, severity, and location).

c. **Incident Lifecycle Manager:** Tracks events from the moment of detection through supervisor acknowledgement to final resolution.

d. **Analytics Engine:** Aggregates ergonomic and safety data into trend-based visual reports for long-term planning.

4.3 Design Methodology

The project utilizes an **Iterative, User-Centered Design (UCD) methodology**. This approach is essential for industrial safety systems where cognitive overload can lead to critical errors. The process involves repeated cycles of prototyping and refinement focusing on:

Information hierarchy and severity-based color coding.

Interface readability for control-room distances.

Mobile interaction efficiency for users in high-pressure field conditions.

Error prevention during high-impact actions like incident closure.

4.4 Detailed Component Design

The project design is modular, where each module corresponds to a distinct user task and interaction context:

Monitoring Module (Web-centric):

Purpose: Enables continuous scanning across multiple camera feeds.

Design Details: Features a stable grid layout, clear camera labeling, and a ”Focus Mode” for high-priority streams.

Emergency Incident Module (Dual-platform):

Purpose: Surfaces urgent hazards requiring human intervention.

Design Details: Uses prominent visual cues (red/amber badges) and provides immediate access to incident snapshots.

Incident Detail & Evidence Module:

Purpose: Supports quick confirmation and documentation of safety events.

Design Details: Displays anonymized evidence images, structured action logs, and a chronological timeline of the event lifecycle.

Ergonomics Module (Preventive Analytics):

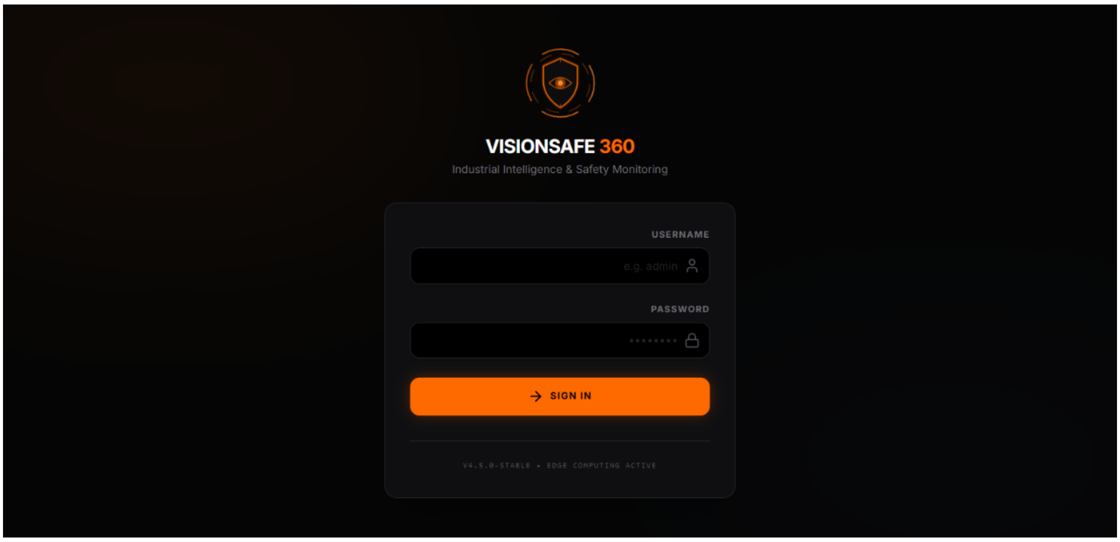
Purpose: Supports the reduction of long-term musculoskeletal risks.

Design Details: Utilizes trend charts, exposure indicators, and zone-based heatmaps rather than intrusive alarms.

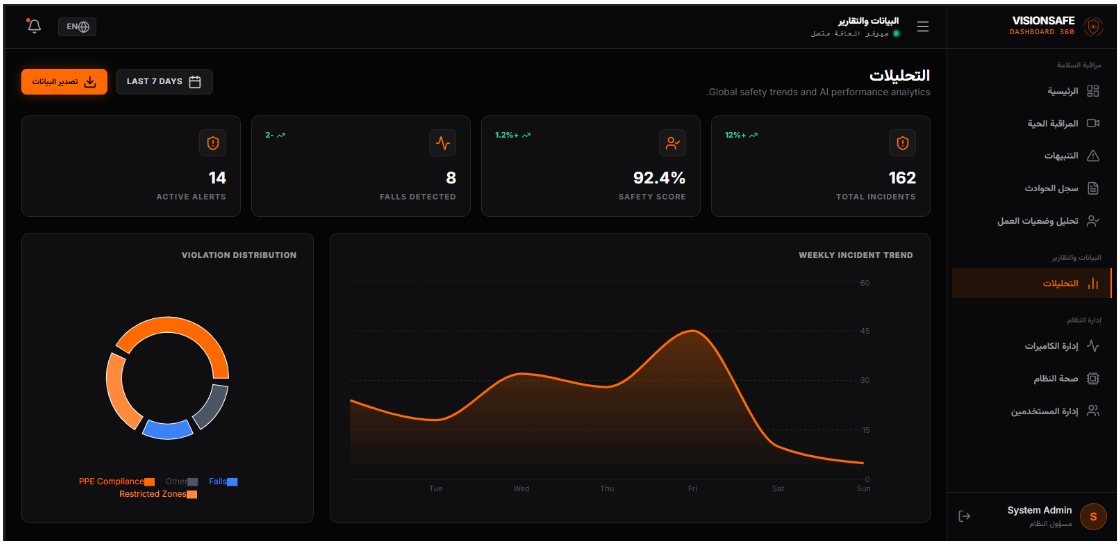
4.5 UI/UX Design

A. Overview

VisionSafe 360 delivers a unified experience where the **Web Dashboard** is optimized for investigation and the **Mobile App** is optimized for action. The design ensures consistent terminology, severity language, and navigation logic across both platforms.




VISIONSAFE
DASHBOARD 360
DASHBOARD
حالة سلامة المصنع
32/34
1.2h
45s
12
اتجاهات السلامة (أسبوعي)
المناطق عالية الخطورة
High
Medium
High
Tue
Wed
Thu
Fri
Sat
Sun
System Admin


VISIONSAFE
DASHBOARD 360
البيانات والتقارير
التحليلات
LAST 7 DAYS
14
ACTIVE ALERTS
8
FALLS DETECTED
92.4%
SAFETY SCORE
162
TOTAL INCIDENTS
VIOLATION DISTRIBUTION
WEEKLY INCIDENT TREND
PPE Compliance
Other
Falls
Restricted Zones
System Admin

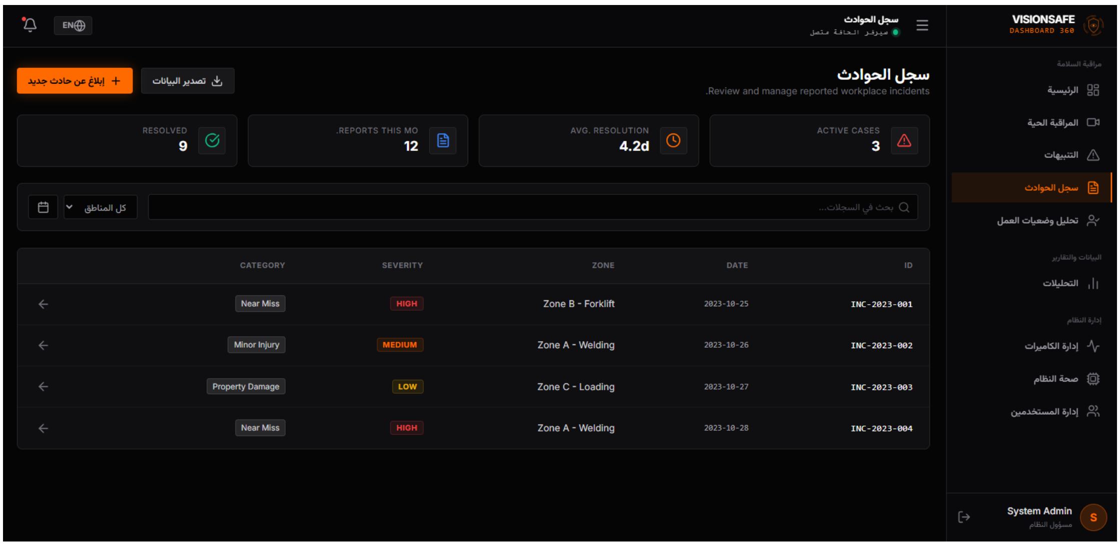

VISIONSAFE
DASHBOARD 360
سجل الحوادث
Review and manage reported workplace incidents.
RESOLVED
9
REPORTS THIS MO.
12
AVG. RESOLUTION
4.2d
ACTIVE CASES
3
CATEGORY
SEVERITY
ZONE
DATE
ID
Near Miss
HIGH
Zone B - Forklift
2023-10-25
INC-2023-001
Minor Injury
MEDIUM
Zone A - Welding
2023-10-26
INC-2023-002
Property Damage
LOW
Zone C - Loading
2023-10-27
INC-2023-003
Near Miss
HIGH
Zone A - Welding
2023-10-28
INC-2023-004
System Admin

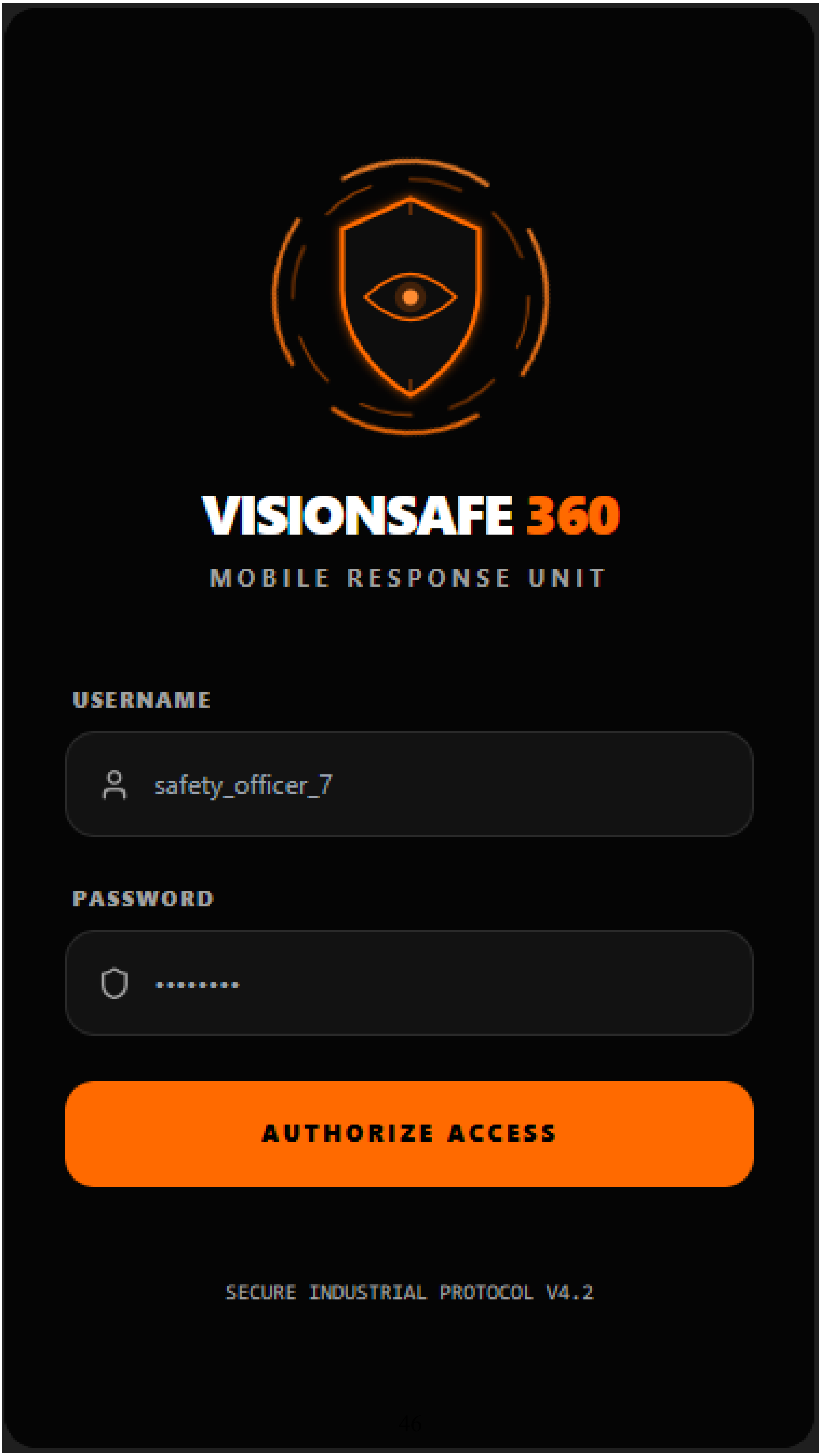
Figure [17]: Web Dashboard Main Interface Layout

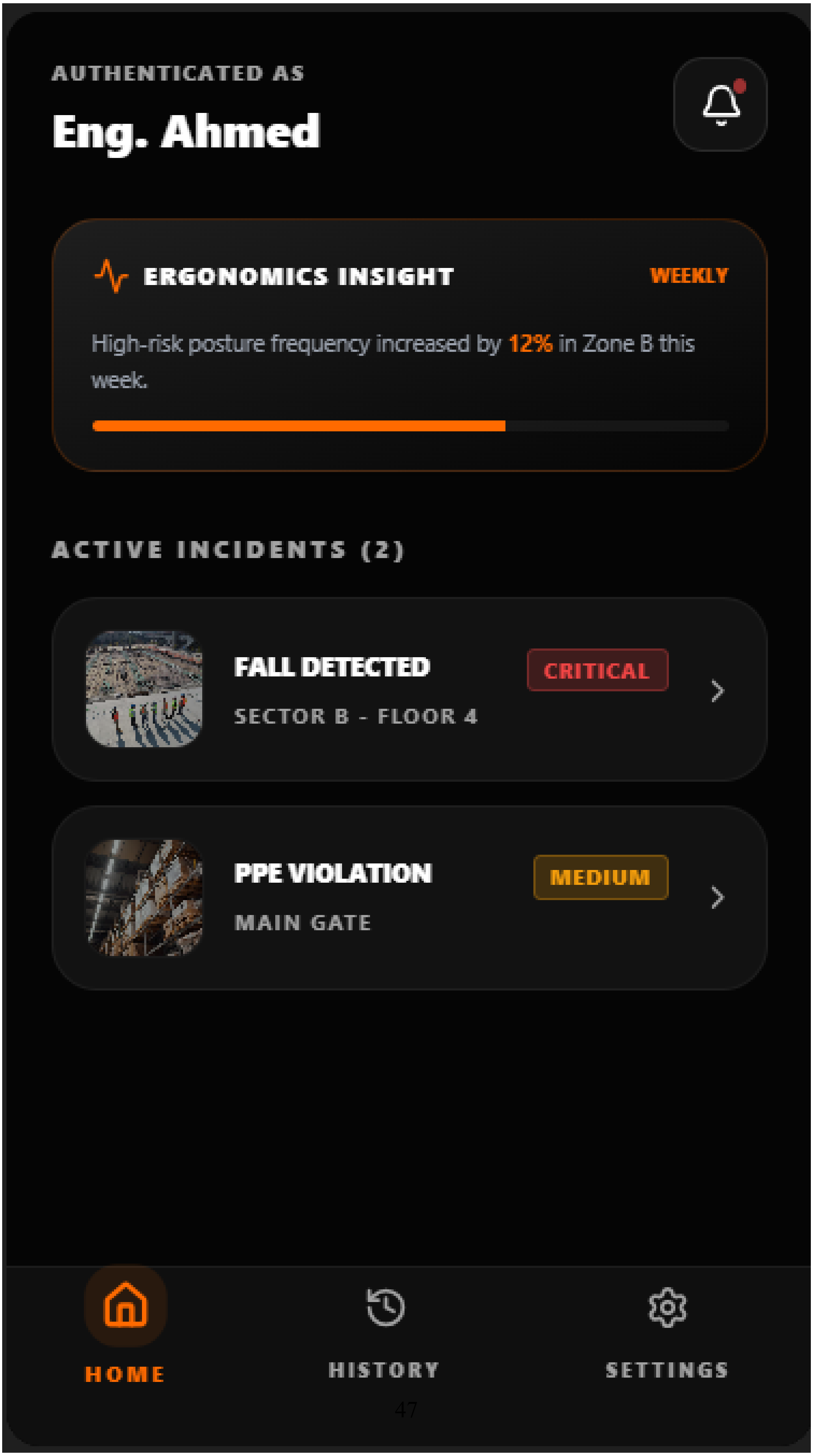
INITIALIZING SECURE PROTOCOL...

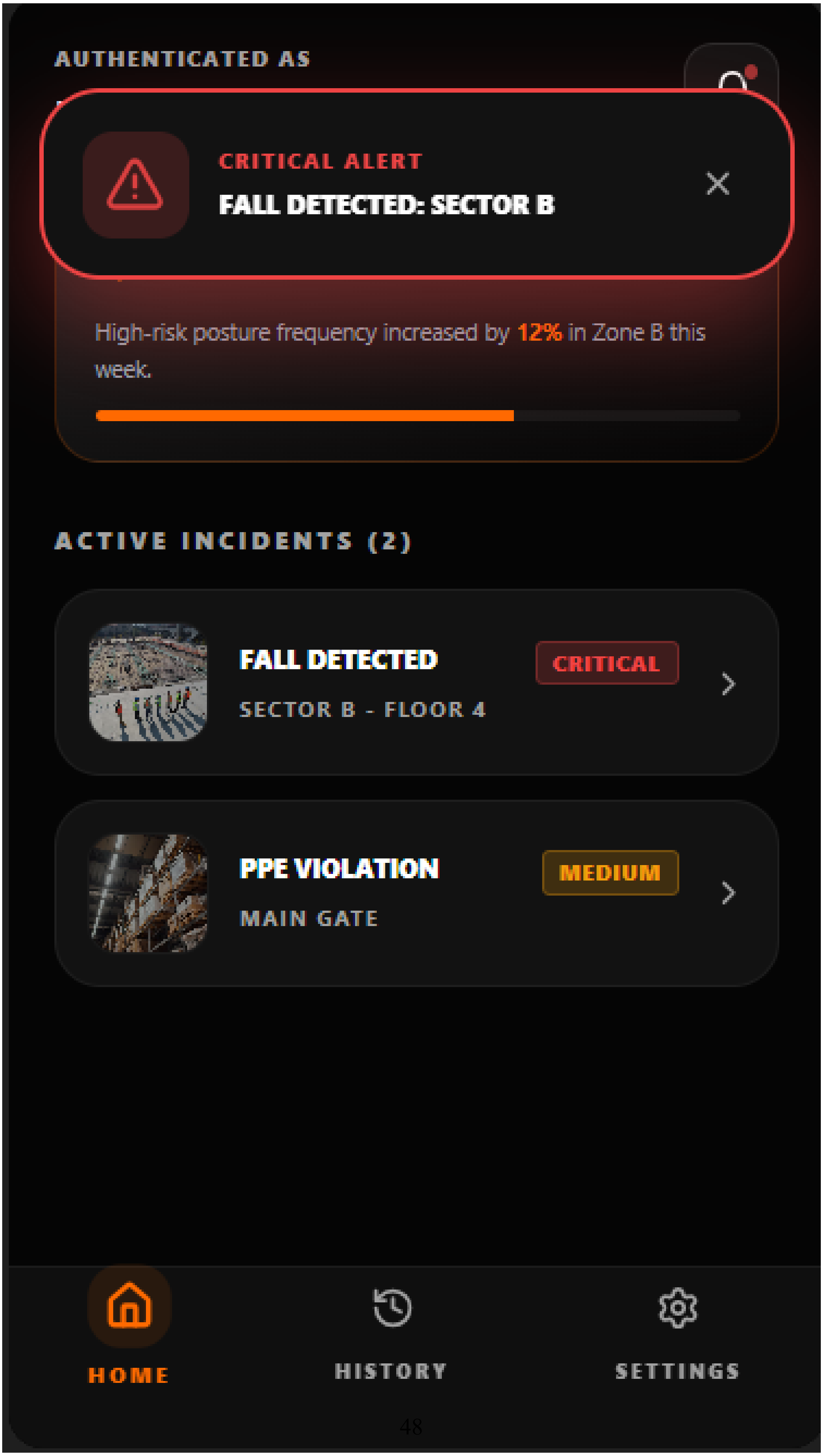
SECURE_HANDSHAKE_V4.2... OK

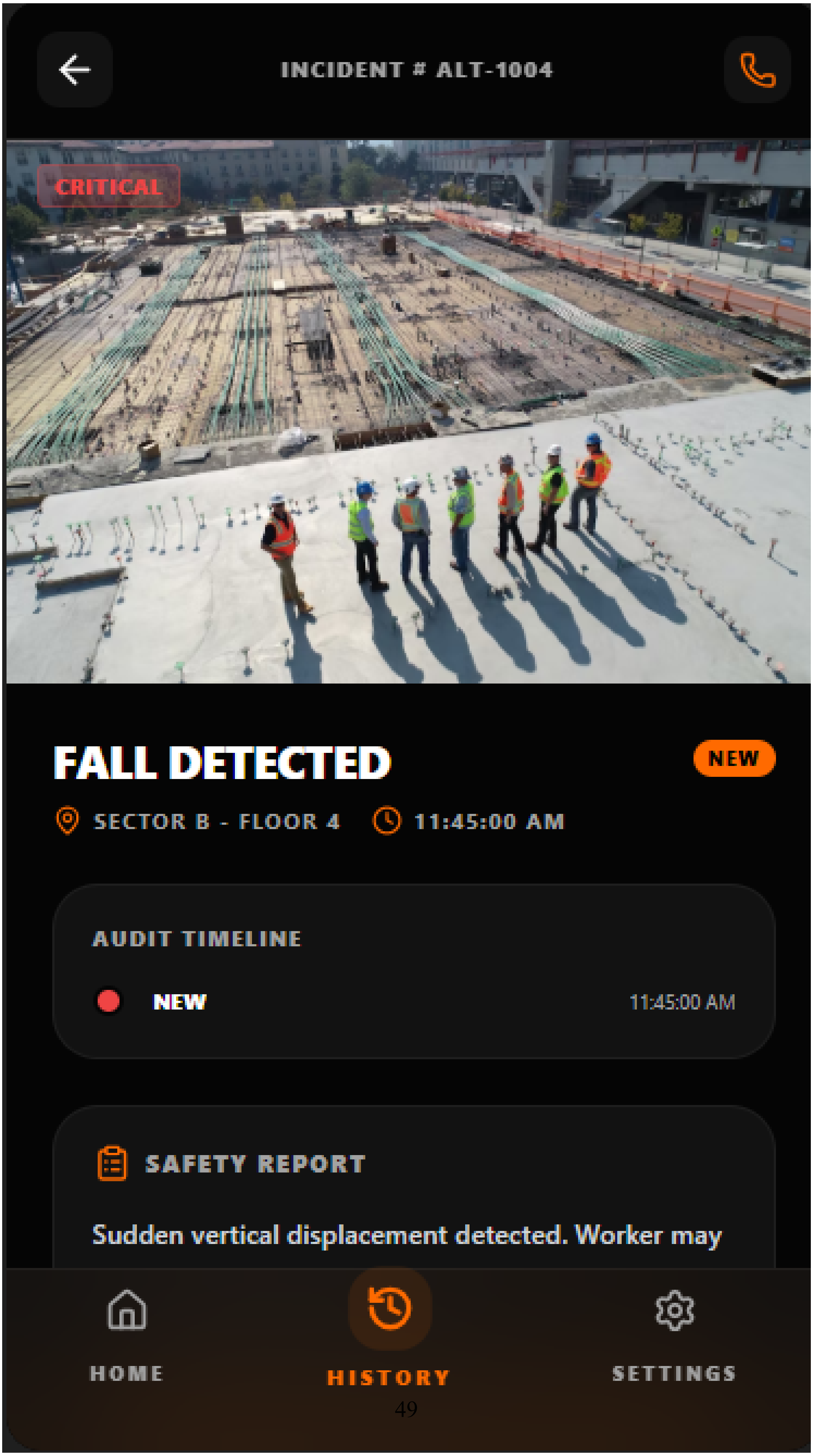
34%

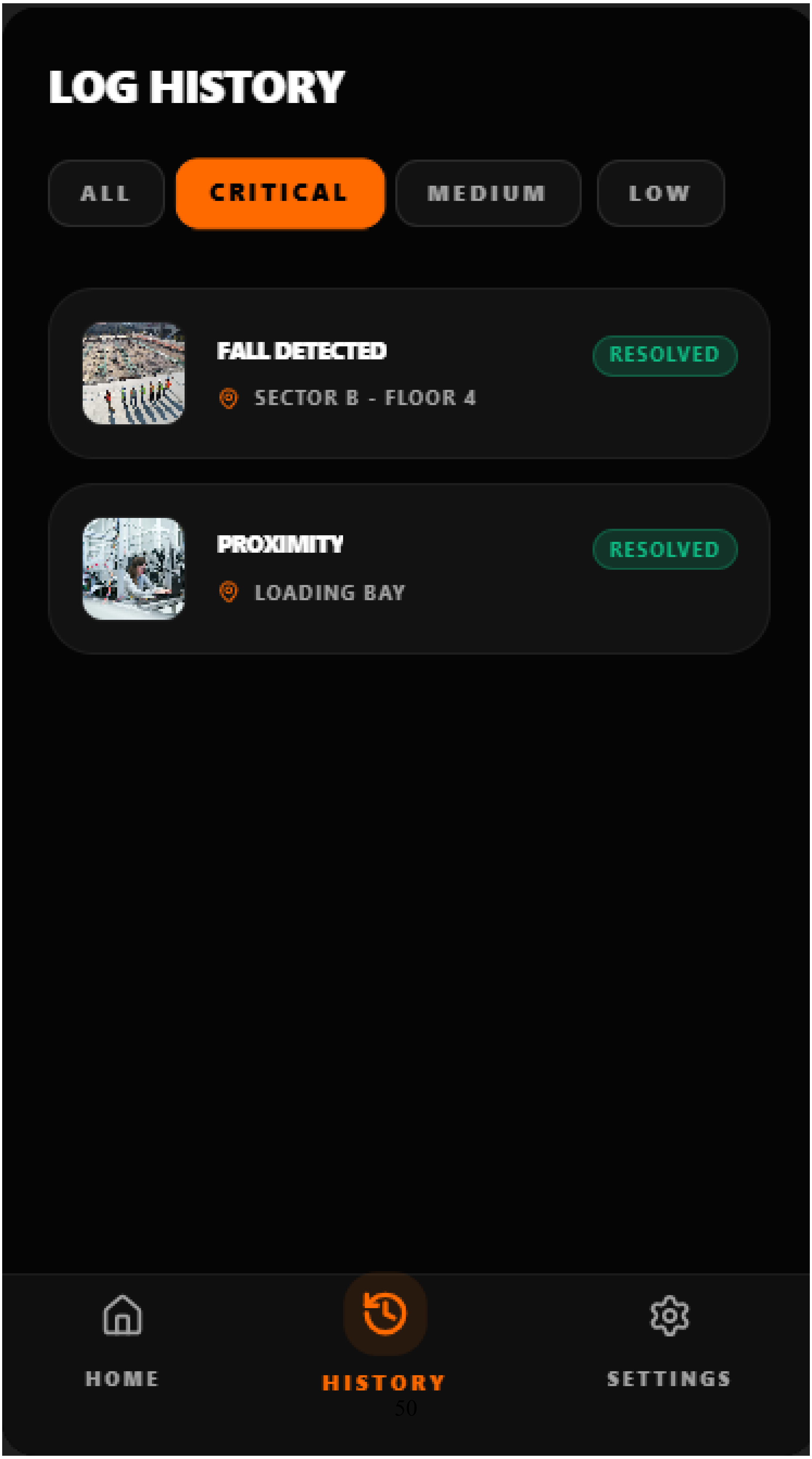
VISIONSAFE ENTERPRISE

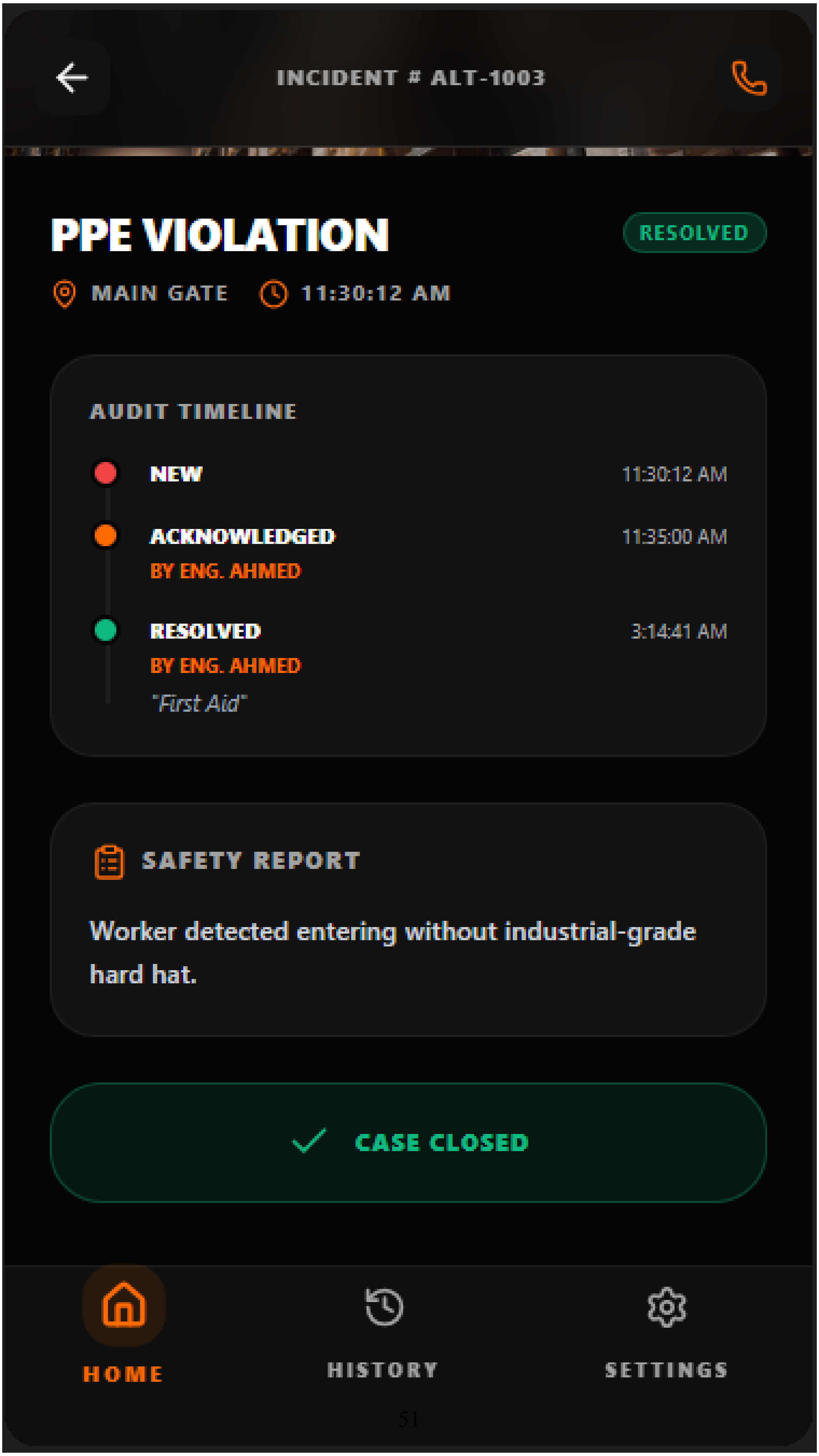

VISIONSAFE 360
MOBILE RESPONSE UNIT
USERNAME
safety_officer_7
PASSWORD
••••••••
AUTHORIZE ACCESS
SECURE INDUSTRIAL PROTOCOL V4.2

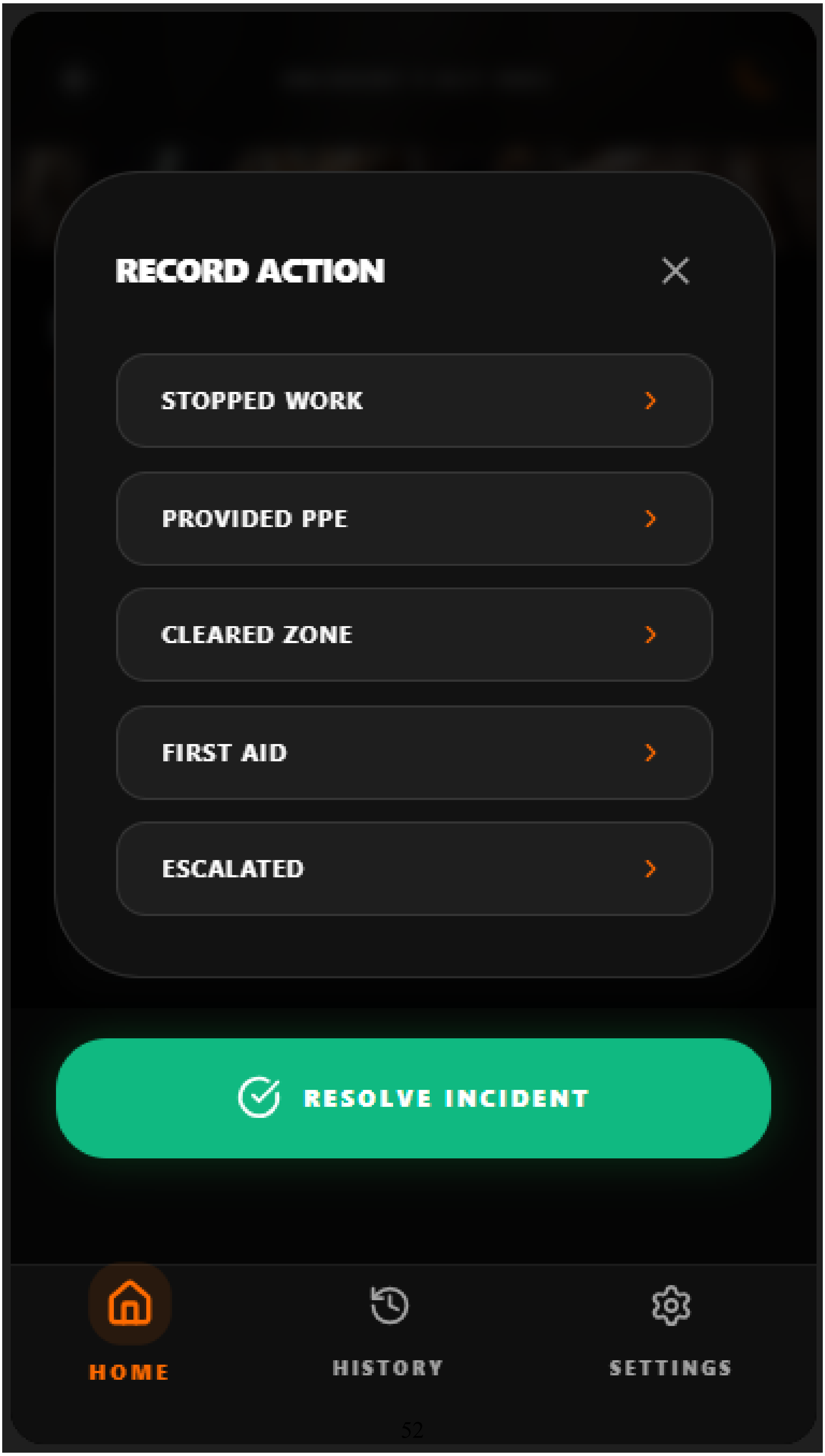

AUTHENTICATED AS
Eng. Ahmed
ERGONOMICS INSIGHT
WEEKLY
High-risk posture frequency increased by 12% in Zone B this week.
ACTIVE INCIDENTS (2)
FALL DETECTED
CRITICAL
SECTOR B - FLOOR 4
PPE VIOLATION
MEDIUM
MAIN GATE
HOME
HISTORY
SETTINGS


AUTHENTICATED AS
CRITICAL ALERT
FALL DETECTED: SECTOR B
High-risk posture frequency increased by 12% in Zone B this week.
ACTIVE INCIDENTS (2)
FALL DETECTED
CRITICAL
SECTOR B - FLOOR 4
PPE VIOLATION
MEDIUM
MAIN GATE
HOME
HISTORY
SETTINGS


INCIDENT # ALT-1004
CRITICAL
FALL DETECTED
NEW
SECTOR B - FLOOR 4
11:45:00 AM
AUDIT TIMELINE
NEW
11:45:00 AM
SAFETY REPORT
Sudden vertical displacement detected. Worker may
HOME
HISTORY
SETTINGS


LOG HISTORY
ALL
CRITICAL
MEDIUM
LOW
FALL DETECTED
RESOLVED
SECTOR B - FLOOR 4
PROXIMITY
RESOLVED
LOADING BAY
HOME
HISTORY
SETTINGS


INCIDENT # ALT-1003
PPE VIOLATION
RESOLVED
MAIN GATE
11:30:12 AM
AUDIT TIMELINE
NEW
11:30:12 AM
ACKNOWLEDGED
BY ENG. AHMED
11:35:00 AM
RESOLVED
BY ENG. AHMED
"First Aid"
3:14:41 AM
SAFETY REPORT
Worker detected entering without industrial-grade hard hat.
CASE CLOSED
HOME
HISTORY
SETTINGS


RECORD ACTION
STOPPED WORK
PROVIDED PPE
CLEARED ZONE
FIRST AID
ESCALATED
RESOLVE INCIDENT
HOME
HISTORY
SETTINGS

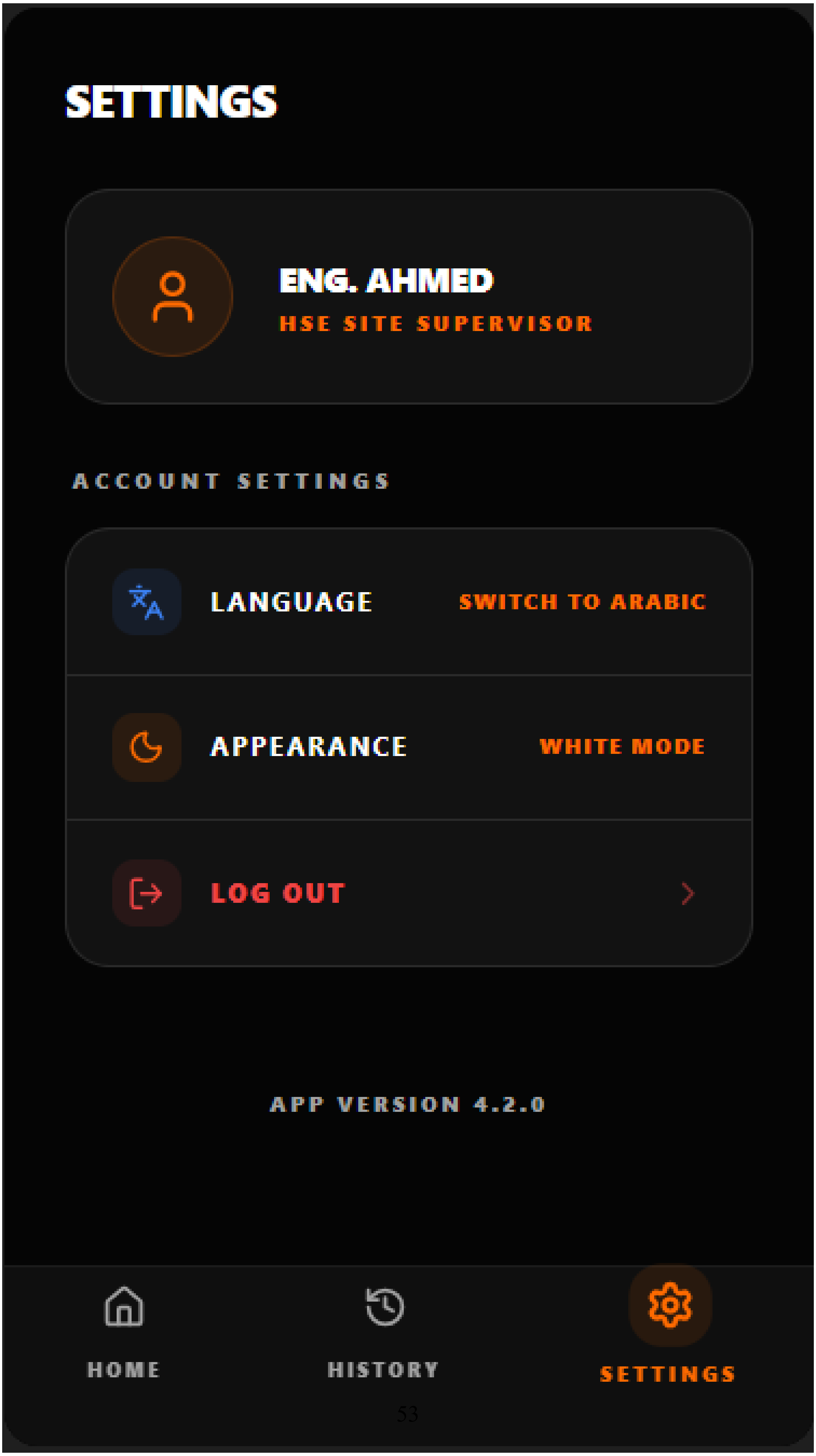

SETTINGS
ENG. AHMED
HSE SITE SUPERVISOR
ACCOUNT SETTINGS
LANGUAGE
SWITCH TO ARABIC
APPEARANCE
WHITE MODE
LOG OUT
APP VERSION 4.2.0
HOME
HISTORY
SETTINGS

Figure [18]: Mobile Application Interface for Field Alerts

B. User Interaction

The application ecosystem allows users to:

Monitor multiple camera views and quickly focus on a specific feed.

Review emergency incidents with clear severity and location context.

Acknowledge incidents via mobile or web to confirm awareness and ownership.

Resolve incidents by recording corrective actions for audit purposes.

Outputs are delivered through:

Incident Cards: Summarizing type, severity, location, and timestamp.

Visual Evidence: Anonymized snapshots for rapid incident confirmation.

Trend Dashboards: Summary indicators and charts for long-term analysis.

C. Design Principles

Simplicity: Reducing cognitive load through clean layouts and minimal steps for critical actions.

Clarity and Hierarchy: Prioritizing urgent hazards while keeping secondary details available but not dominant.

Accessibility: High contrast, readable typography, and non-color-only severity cues (using icons and text).

Error Prevention: Confirmation dialogs for destructive actions and clear system-state indicators (e.g., offline camera).

4.6 Workflow Design

A. Emergency Incident Workflow

This workflow ensures that every detected hazard is accounted for by a human supervisor.

Detection: AI identifies a hazard and triggers an alert.

Notification: Alert appears on the Web Dashboard and the Mobile App simultaneously.

Acknowledgement: A supervisor confirms ownership via the mobile device.

Resolution: The supervisor records the corrective action taken to close the incident.

B. Ergonomics Workflow

Passive Monitoring: Ergonomic risks are logged as data points without immediate alarms.

Trend Analysis: HSE Managers review cumulative data on the Web Dashboard weekly/monthly.

Strategic Intervention: Decisions are made based on high-risk zones or recurring posture issues.

4.7 Database Design (Conceptual)

A. Overview

The database is structured to support the core UX requirements: traceability, accountability, and reporting. It ensures that data remains consistent across the Web and Mobile interfaces.

B. Table Descriptions

Users & Roles: Manages identity and access scope (Admin, Manager, Supervisor).

Cameras: Stores camera metadata and location tagging for UI labeling.

Incidents: Core table tracking hazard types, severity, snapshots, and lifecycle status.

Ergonomic Records: Stores posture risk summaries and exposure indicators for analytics.

Action Logs: Maintains an audit trail for every acknowledgement and resolution action.

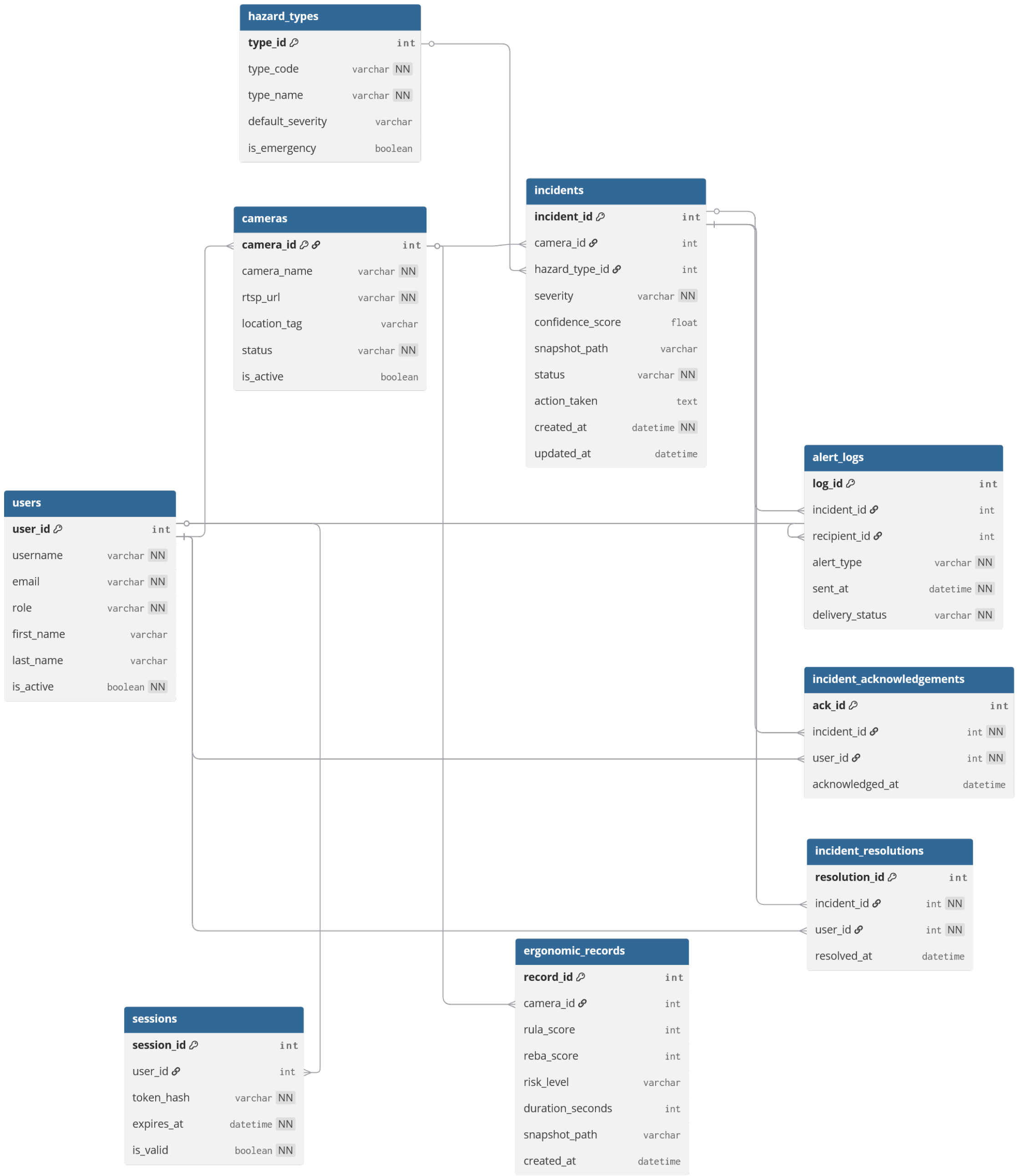


Figure [19]: Entity-Relationship Diagram (ERD)

4.8 Design Constraints

a. **Operational Context:** Industrial environments introduce noise and distractions, requiring high-visibility UI.

b. **Monitoring Duration:** Visual design must minimize eye strain for control-room operators during long shifts.

c. **Device Diversity:** The design must remain usable on large desktop displays and ruggedized mobile screens.

d. **Privacy Requirements:** Evidence presentation must respect worker privacy through anonymization techniques.

4.9 Rationale for Design Choices

a. **Performance (UX):** The interface is structured to surface the most critical information first, reducing time-to-action.

b. **Platform Synergy:** Using the Web for ”Heavy Data” (Analytics) and Mobile for ”Fast Action” (Alerts) aligns with the natural professional behavior of safety officers.

c. **Scalability:** Consistent navigation and layout patterns remain usable as the number of monitored cameras increases.

4.10 Integration of Avatars and Gesture Translation

This section is not applicable to VisionSafe 360. The project scope is focused on industrial safety monitoring and incident workflows and does not involve avatar-based sign rendering or gesture translation.

4.11 Backend and Application Development

The application layer ensures that the user experience remains coherent across devices. The **Web Dashboard** is developed as a command hub for deep investigation, while the **Mobile Application** is designed as a field-oriented response tool. Both platforms communicate through a centralized backend that ensures real-time updates, predictable status transitions, and unified records, minimizing the need for manual synchronization or extensive operator training.